

School of Computing and Mathematical Sciences

CO7201 Individual Project

Final Report

An Online Tool to Manage Pension Benefits

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Abstract

Managing pension benefits is often a complex and fragmented process, particularly within the framework of workplace pension schemes in the United Kingdom. This project presents the design and development of a secure, scalable, and user-friendly digital platform that enables individuals to effectively manage their pension accounts. The system emulates a modern workplace pension scheme similar to those offered by "NEST" and "The People's Pension" by allowing users to view monthly contributions, track fund performance, manage personal and beneficiary details, and download official pension statements. An interactive dashboard provides real-time visualization of pension growth, empowering users with greater financial transparency.

Robust security measures were a central focus throughout development, including the implementation of multi-factor authentication (MFA), role-based access control (RBAC), secure password hashing, and encrypted communication. These features ensure compliance with General Data Protection Regulation (GDPR) and pension governance standards set by The Pensions Regulator (TPR).

The platform adopts a three-tier architecture comprising a dynamic Angular frontend, a Flask-powered RESTful backend, and a MySQL relational database for persistent, structured data storage. Key features include automated monthly statement generation, PDF export, document upload functionality, and chatbot integration for user assistance. The system is modular and extensible, allowing future integration of AI-driven retirement forecasting tools, gamification elements, and multilingual support.

This report documents the complete software development lifecycle ,covering requirements gathering, system design, technical implementation, testing strategy, evaluation, and critical comparison with existing pension platforms. It demonstrates how advanced software engineering techniques can be applied to address real-world pension management challenges while meeting compliance, usability, and security objectives.

Contents

1. INTRODUCTION 4

1.1 Background 4
1.2 Motivation 5
1.3 Project Aim 5
1.4 Objectives 6

2. REQUIREMENT SPECIFICATION 8

2.1Functional Requirements 8
2.2 Non-functional Requirements 10
2.3 Essential Requirements 11
2.4 Desirable Requirements 12
2.5 Optional Requirements 12

3. LITERATURE REVIEW 13

3.1 Security and Authentication 13
3.2 Pension Management and Governance 14
3.3 Cloud Computing for Financial Applications 15
3.4 User Experience (UX) for Digital Financial Platforms 16
3.5 Emerging Technologies: AI, Automation, and Gamification 17
3.6 Comparison Between the Proposed System and Existing Systems 18

4. SYSTEM DESIGN 19

4.1 System Architecture 19
4.2 Pension Provider Simulation 21
4.3 Statement Structure 23
4.4 Use Case Diagram 26
4.5 Database Design 28
4.6 Technology Stack Justification 31

5. IMPLEMENTATION 33

5.1 Frontend Development 33
5.2 Backend Development 35
5.3 Database Implementation 37
5.4 Highlighted Code Areas 39

6. TESTING AND EVALUATION 47

6.1 Postman API Testing 47
6.2 Integration Testing – Postman 51
6.3 Functional Testing – Postman 51
6.4 Security Testing – Postman 52
6.5 Internationalization (i18n) Testing 52
6.6 Manual UI Testing 55

7. CRITICAL EVALUATION 57

7.1 Achievements 57
7.2 Challenges and Solutions 58

8. FUTURE WORK 62

9. CONCLUSION 64

10. REFERENCES 65

1. Introduction

1.1 Background

Pension management is a critical aspect of financial planning, providing individuals with a secure income during retirement. Traditionally, pension information has been managed through paper-based systems, periodic statements sent by post, or fragmented online portals. These methods often result in limited accessibility, poor transparency, and difficulties in tracking long-term contributions and fund performance. With the rise of digital financial services and the growing expectations for real-time access to personal financial data, there is a pressing need to modernize pension management platforms.

In the United Kingdom, workplace pension schemes have become mandatory under auto-enrolment laws, requiring employers to contribute a percentage of employees' salaries towards pension plans. Major providers such as "NEST" and "The People's Pension" have set standards by offering digital access to pension accounts, monthly contribution tracking, and downloadable statements. However, even among leading platforms, challenges persist in delivering highly secure, user-centric, and scalable solutions that comply with regulatory frameworks like the General Data Protection Regulation (GDPR) and guidelines from The Pensions Regulator (TPR).

Given the sensitive nature of pension data and the diversity of users including employees, employers, and administrators any robust pension platform must carefully balance security, usability, scalability, and compliance. The backend must safeguard personal and financial data, while the user interface must accommodate individuals with varying levels of digital literacy.

This project seeks to meet these expectations by building a secure and interactive online platform for managing workplace pensions. The system offers real-time contribution tracking, fund performance visualization, downloadable statements, personal and beneficiary profile updates, and a responsive dashboard experience. Employers can enroll Employee, while administrators can manage user roles and monitor system-wide activity, contribution data.

Developed using a three-tier architecture comprising an Angular frontend, Flask REST API backend, and MySQL database the project simulates a modern workplace pension environment. It also lays the groundwork for future extensions such as AI-assisted financial planning, multilingual access. By bridging operational needs with regulatory and technical standards, the platform contributes to the ongoing evolution of digital pension services in the UK.

1.2 Motivation

The motivation for this project arises from the growing demand for secure, accessible, and user-friendly pension management solutions. Despite advancements in financial technologies, many users still face fragmented access to their pension information, confusing contribution records, and limited real-time visibility into their retirement savings. These challenges can lead to disengagement with pension planning, ultimately affecting long-term financial security.

Workplace pension schemes in the United Kingdom, such as those operated by "NEST" and "The People's Pension", have made significant strides toward digital transformation. However, gaps remain, particularly in offering intuitive platforms that combine strong security measures with easy-to-use interfaces, real-time contribution tracking, and comprehensive fund insights. Furthermore, compliance with evolving regulations such as GDPR places additional requirements on pension systems to safeguard sensitive personal and financial data.

This project is motivated by an opportunity to address these shortcomings through the development of a modern, simulated pension management system that empowers not only employees but also employers and administrators. For employers, the system streamlines employee management. For admins, contribution uploads and it offers tools for oversight, compliance monitoring, and system health. For employees, it provides secure login, real-time pension visibility, personalized fund analytics, and downloadable monthly statements all backed by MFA, RBAC, and encrypted data handling.

Beyond technical problem-solving, this project also reflects a deeper academic and personal motivation to apply advanced full-stack development skills to a socially meaningful domain. In an era where digital self-service and financial literacy are increasingly vital, building a secure, modular, and extensible pension platform offers both practical value and academic challenge.

Ultimately, this work aims to demonstrate that with thoughtful system design and user-centred development, digital pension tools can move beyond mere compliance to foster genuine engagement, trust, and proactive retirement planning.

1.3 Project Aim

The primary aim of this project is to design and develop a secure, scalable, and user-friendly online platform that enables individuals to manage their pension benefits efficiently. The system specifically simulates a UK workplace pension scheme model, similar to providers like "NEST" and "The People's Pension", offering users real-time access to their contribution history, pension fund performance, and downloadable pension statements.

In response to the sensitive nature of financial and personal information, the platform places a strong emphasis on data security and regulatory compliance. Key measures include multi-factor authentication (MFA), role-based access control (RBAC), hashed password storage using Flask-Bcrypt, and encrypted data transmission through HTTPS/TLS protocols. These

safeguards align with legal frameworks such as the General Data Protection Regulation (GDPR) and guidelines from The Pensions Regulator (TPR).

The system is designed to be intuitive and responsive, featuring a modern frontend built with Angular and an efficient, modular backend powered by Flask. Data is managed via a secure MySQL relational database, supporting robust, scalable data storage and query performance. Features include interactive pension analytics with Chart.js, document upload and access controls, and support for multiple user roles namely employees, employers, and administrators.

Additionally, the platform serves as a demonstrative prototype, showcasing how contemporary full-stack development techniques can be leveraged to meet real-world pension governance challenges. It sets the foundation for future enhancements such as AI-assisted retirement forecasting, multilingual accessibility, and gamified user engagement. Through this project, the aim is to demonstrate not only technical proficiency in secure system development, but also the importance of user-centred design, regulatory alignment, and scalability in modern digital financial service applications.

1.4 Objectives

The objectives of this project focus on developing a secure, compliant, and user-friendly pension management system that simulates the operational model of a typical UK workplace pension scheme. The following key objectives guided the project development across technical, functional, and regulatory dimensions:

1. Secure User Authentication and Access Control
- Implement secure login functionality using multi-factor authentication (MFA) to protect user accounts.
- Apply role-based access control (RBAC) to define permissions for different user types (e.g., employee, employer, admin).

2. Pension Contribution Tracking
- Enable users to view monthly contributions made by both the employer and employee.
- Provide a clear breakdown of contributions, contribution dates, and fund performance within the dashboard.

3. Personal Information Management
- Allow users to securely view and update their personal details through the portal.
- Enable beneficiary management, allowing users to add/edit beneficiaries and define fund allocation percentages.

4. Pension Statement Management
- Automatically generate monthly pension statements summarizing contributions, fund performance, and pot value.
- Provide functionality to download pension statements securely in PDF format.

5. Fund Visualization and Analytics
- Integrate visual charts (using Chart.js) to display the growth of the pension pot over time.
- Offer basic projections of fund growth based on standard performance assumptions.

6. Administrative and Employer Functions

- Allow employers to add employees, upload monthly contributions, and view/manage employee data and statements.
- Enable administrators to manage users, employers, and system-wide pension data.
- Support functions such as generating or deleting statements and viewing nested employer-employee relationships.

7.System Security and Compliance

- Ensure all data storage and adhere to GDPR standards and best practices in financial application security.
- Secure backend APIs with encrypted communication (HTTPS/TLS).
- Store passwords using Flask-Bcrypt hashing and implement secure token-based authentication with JWT.

8. Extended Functionalities Implemented

- Integrate a Voiceflow-based chatbot assistant to help users with pension-related queries.
- Provide multi-language UI support via ngx-translate, currently supporting English and Hindi, Polish, French, German with the structure in place to support additional languages.

9. Testing and Validation

- Perform API-level testing using Postman and manual validation across all major user flows.
- Implement input validation and error handling to maintain data integrity and improve system reliability.

Each objective has been carefully designed to address real-world pension management challenges, align with user expectations for digital financial services, and ensure that the system can serve as a foundation for future improvements and extensions.

2. Requirement Specification

The requirements for the pension management system were gathered using a hybrid methodology combining document analysis, stakeholder persona mapping, and benchmarking of real-world platforms like NEST and The People's Pension. Key regulations and standards from The Pensions Regulator (TPR) and GDPR were analysed to ensure legal compliance, especially around auto-enrolment, secure data handling, and user consent. Role-specific needs for employees, employers, and administrators were defined through persona-based use cases, identifying features such as pension statement access, contribution uploads, and system oversight. Comparative analysis helped incorporate modern enhancements like chatbot support (Voiceflow), pension analytics (Chart.js), and multilingual UI (ngx-translate). Requirements were iteratively refined through Postman API testing, UI validation, and role-based access mapping, ensuring the final system was secure, scalable, and aligned with both regulatory and user-centred design principles.

2.1Functional Requirements

The development of the pension management system focuses on delivering core functionalities essential for users, administrators, and employers to interact with the platform securely and efficiently. The following functional requirements have been identified and prioritized to meet the objectives of the project:

1. Secure User Authentication

- Users must be able to log in securely with password and multi-factor authentication (MFA).
- Clear feedback must be provided for login errors (e.g., invalid credentials, MFA failure).
- Sessions must be maintained using JWT-based authentication.

2. Role-Based Access Control (RBAC)

- The system must define and enforce three distinct roles: Employee, Employer, and Admin.
- Access permissions must be scoped:
    - **Employees**: access only their data and statements.
    - **Employers**: manage and view their linked employees.
    - **Admins**: manage all users, employers, and statements.

3. Pension Statement Viewing

- Employees must be able to view detailed monthly pension statements showing:
    - Employer and employee contributions.
    - Total monthly contribution.
    - Fund performance and pension pot value.
- Statements must be displayed in a clear, tabular layout.

4. Pension Statement Download

- Employees must be able to securely download pension statements in PDF format.
- PDF layout must match dashboard data and include disclaimers for confidentiality and security.

5. Personal Information and Beneficiary Management

- Employees must be able to view and update personal details such as salary, NI number, and address.
- The platform must allow users to add/edit beneficiaries and assign allocation percentages.

6. Document Management

- Employees must be able to upload documents (e.g., proof of ID, contracts) in secure formats (.pdf, .jpg, .png).
- The system must allow users to view, download, and organize uploaded documents.

7. Contribution Management (Employer/Admin)

- Employers and Admins must be able to:
  - Upload monthly contribution data for their employees.
  - Edit or update contribution records when necessary.
  - View complete contribution histories for assigned users.

8. Fund Visualization

- Employees must be able to visualize their pension pot growth using interactive charts (Chart.js).
- Charts must dynamically reflect changes and be responsive across devices.

9. Security Enforcement

- All passwords must be securely hashed using Flask-Bcrypt.
- All API communications must occur over HTTPS/TLS.
- Endpoints must enforce secure access using JWT-based session tokens.

10. Admin Dashboard and Oversight

- Admins must be able to:
  - Add/delete employers.
  - View and manage all users across the platform.
  - Generate or delete employee pension statements.
  - View employer-employee associations.

11. AI Chatbot Integration

- Embed a Voiceflow-powered AI chatbot to answer pension-related questions for users.

12. Multilingual UI Support

- Interface content must be dynamically translatable using ngx-translate.
- English, Polish, French, German and Hindi must be fully supported, with provisions to add languages.

13. Notifications

- The system must notify employees when a new pension statement has been generated for them.

2.2 Non-functional Requirements

In addition to fulfilling functional specifications, the pension management system was designed to meet a comprehensive set of non-functional requirements. These characteristics ensure that the platform is secure, performant, scalable, and aligned with real-world expectations for usability and compliance in digital financial services.

1. Security

- All data exchanges between the Angular frontend and Flask backend must be secured using HTTPS/TLS.
- Sensitive data, including user credentials and personal details, must be encrypted and securely stored.
- JWT (JSON Web Token) must be implemented to manage user sessions securely and validate every API call.
- Multi-factor authentication (MFA) must be enforced for privileged roles such as employers and administrators.
- Role-based access control (RBAC) must limit access to system features based on user type.

2. Usability

- The user interface must be intuitive and easy to navigate for users with varying levels of technical experience.
- Workflows such as login, statement viewing, and profile updates must be logical and user-friendly.
- The UI must be responsive across desktop, tablet, and mobile devices.
- Multi-language support must be available to enhance accessibility for diverse user bases (e.g., English, Hindi, Polish, French, German).
- An embedded chatbot (Voiceflow) must assist users by answering queries and guiding them through key actions.

3. Scalability

- The system must handle increasing volumes of users and pension data without performance degradation.
- The backend architecture must support modular scaling (e.g., horizontal scaling of the Flask app or distributed database hosting if needed).

4. Performance

- Webpages must load within a maximum of 3 seconds under normal operating conditions.
- API endpoints must respond to typical user requests (e.g., viewing statements, submitting updates) within 2 seconds.
- Background tasks, such as statement PDF generation, must complete without causing noticeable delays to the user.

5. Maintainability

- The platform must follow a modular architecture separating frontend, backend, and database layers.
- Source code must be well-commented

- and follow consistent naming conventions.
- GitLab must be used for version control, enabling collaborative development, feature branching, and rollback.
- Manual and API-level testing (e.g., via Postman) must be conducted to validate system functionality and minimize regression bugs.

6. Availability and Reliability

- Regular automated database backups must be scheduled to prevent data loss.
- The system must provide basic error handling, fallback messages, and page routing during network or backend failures.
- Token expiration and session validation must prevent unauthorized long-term access and handle timeout scenarios gracefully.

7. Regulatory Compliance

- The system must simulate compliance with The Pensions Regulator (TPR) standards regarding contribution tracking and record maintenance.
- It must also ensure simulated compliance with GDPR by protecting personal data and allowing users to update or delete their information upon request.
- All stored documents and downloadable statements must include disclaimers and adhere to best practices in financial recordkeeping.

2.3 Essential Requirements

These features are mandatory for the system to operate as a secure, compliant, and user-focused pension management platform.

1. **Secure Authentication and User Management**
    - Users must log in using securely hashed credentials with multi-factor authentication (MFA) for added protection.
    - JWT-based session management and token expiration enforcement.
2. **Pension Statement and Contribution History Tracking**
    - Auto-generate and display monthly pension statements with employer/employee contributions, pot value, and fund performance.
3. **Personal Details and Beneficiary Management**
    - Allow users to securely view and update personal details.
    - Enable users to manage beneficiary records with relationship and percentage allocation.
4. **Role-Based Access Control (RBAC)**
    - Implement differentiated access levels for Employees, Employers, and Admins.
    - Employers can manage employee lists and upload contributions.
    - Admins have full control over users, statements, and employers.
5. **Secure API and Database Communication**
    - All sensitive communication must occur over HTTPS/TLS.
    - Passwords must be hashed with Flask-Bcrypt; endpoints must use JWT-based protection.

6. **Pension Statement Download in PDF**
    - Users must be able to download monthly statements in a secure and official PDF format.
7. **Employee Contribution**
    - Admin must be able to upload contribution records for employees via the admin interface.

8. **Admin Panel**
    - Admin must be able to view all users and employers, generate or delete statements, and assign/reset passwords as needed.

2.4 Desirable Requirements

These features enhance usability, engagement, and convenience but are not essential to the system's core functionality.

1. **Real-Time Notifications and Alerts**
    - Notify users when new statements are generated.
2. **Dashboard with Visual Analytics**
    - Use Chart.js to provide pension pot growth visualization and monthly contribution graphs.
3. **Automated Pension Contribution Calculations**
    - System dynamically calculates monthly contributions based on salary, employer/employee percentages (e.g., 3%/5%).
4. **Document Upload and Management**
    - Users can upload, view, and download pension-related documents (e.g., ID proofs, contracts, beneficiary declarations).
5. **Monthly Pension Summary Table**
    - Tabular representation of all statements with quick download buttons and filters by date or period.

2.5 Optional Requirements

These features are not required for initial system delivery but were implemented or scoped for future expansion.

1. **Multi-Language Support**
    - UI supports English, Polish, French, German and Hindi using ngx-translate, with placeholders for future languages.
2. **Chatbot Assistance**
    - Integrated Voiceflow-based chatbot to assist users with pension-related FAQs.
3. **Gamification Elements**
    - Proposed feature: Add motivational badges, savings milestones, or educational nudges to encourage pension engagement.

3. Literature Review

3.1 Security and Authentication

Security is a critical aspect of any online financial platform, particularly one that handles sensitive pension data. The pension management system developed in this project places security at the forefront by implementing a multi-layered approach that includes multi-factor authentication (MFA), role-based access control (RBAC), encrypted data storage, and secure communication protocols. These measures ensure that only authorized users can access pension information, and that personal and financial data are protected against unauthorized disclosure, tampering, or loss.

Multi-factor authentication (MFA) is recognized as one of the most effective methods for securing user accounts. According to the NIST Digital Identity Guidelines [1], incorporating a second authentication factor beyond a password significantly reduces the risk of unauthorized account access, particularly in the event of password compromise. In this project, MFA is implemented as part of the login process to provide an extra layer of verification for users accessing their pension data.

Role-Based Access Control (RBAC) is another critical security mechanism incorporated into the system. Following best practices outlined by the OWASP Top 10 security risks report [12], RBAC ensures that users only have access to data and functionality appropriate to their role within the system. Employees (pension holders) can only view and manage their own information, while administrators (employers) have access to manage employee contributions and statements but not employee credentials or sensitive personal data. This principle of least privilege minimizes the attack surface and helps maintain data segregation.

All sensitive data, including user passwords, is protected using cryptographic hashing algorithms. Passwords are never stored in plaintext; instead, they are securely hashed using Flask-Bcrypt before being persisted in the MySQL database. Communications between the frontend, backend, and database are encrypted using HTTPS and TLS protocols to prevent data interception, tampering, or man-in-the-middle attacks, in alignment with cybersecurity best practices for financial systems [21].

Authentication sessions are managed using JSON Web Tokens (JWT), a stateless, secure method of maintaining user sessions without exposing sensitive information. Upon successful authentication, the backend issues a JWT that is used by the frontend to authorize access to protected resources throughout the user's session.

Incorporating these security measures ensures that the system simulates compliance with the General Data Protection Regulation (GDPR) standards for data protection and privacy [5]. Together, MFA, RBAC, secure password hashing, encrypted communications, and JWT-based session management form a comprehensive security framework, aligning the platform with modern best practices for safeguarding online financial applications [21].

3.2 Pension Management and Governance

Pension management in the United Kingdom operates under a structured and regulated framework designed to ensure transparency, fairness, and long-term financial security for individuals. Governance standards are primarily enforced by bodies such as The Pensions Regulator (TPR), the Financial Conduct Authority (FCA), and HM Revenue and Customs (HMRC). These organizations establish mandatory requirements for workplace pension schemes, including contribution levels, member communications, data security obligations, and regulatory compliance [4], [5].

The Pensions Regulator (TPR) plays a central role in setting out guidelines for workplace pension providers, particularly under the auto-enrolment regime introduced in the UK. Employers are legally obligated to enroll eligible employees into a qualifying pension scheme and contribute a minimum percentage of their qualifying earnings. According to TPR's official guidance [4], the minimum contribution standard is typically set at 5% from employees and 3% from employers, making up a combined total contribution of 8%. Providers must also ensure that members have timely, accurate, and accessible information regarding their pension contributions and fund performance.

The Financial Conduct Authority (FCA) also plays a significant role in pension governance, especially for schemes offering flexible investment options or drawdown facilities. Although this project does not replicate the advisory functions regulated by the FCA, its principles emphasize the necessity for clarity, transparency, and user empowerment within digital pension platforms. Pension management systems are expected to present financial information in a clear, accurate, and non-misleading manner to enable users to make informed retirement planning decisions [5].

Compliance with HMRC requirements is critical in pension management, particularly regarding tax relief on contributions and correct handling of salary deductions. While the developed platform does not integrate directly with HMRC infrastructure, it simulates accurate payroll-based employer and employee contributions, reflecting real-world pension management practices in the UK [20].

Data governance and cybersecurity form additional crucial pillars of pension management. The General Data Protection Regulation (GDPR) mandates that pension providers ensure lawful, fair, and transparent handling of personal financial data, while safeguarding user rights such as data access, correction, and consent withdrawal [21]. In line with these requirements, the developed system incorporates secure user authentication, encrypted storage of sensitive data, strict role-based access control (RBAC), and personal data isolation at the database level.gi

By aligning the platform's design with TPR governance standards [4], FCA transparency guidelines [5], simulated HMRC compliance [20], and GDPR data protection requirements [21], the project not only meets its technical objectives but also models key regulatory practices observed in the real-world pension sector. This ensures that users interact with a system that reflects professional standards of security, transparency, and regulatory adherence.

3.3 Cloud Computing for Financial Applications

Cloud computing has fundamentally transformed the delivery of financial services by enabling scalable, secure, and cost-efficient infrastructures. In the context of pension management platforms, cloud services provide the agility necessary to accommodate large user bases, ensure high availability, and maintain secure storage of sensitive financial data. This project leverages cloud hosting, specifically Firebase, to support the deployment, scalability, and maintenance of the pension management system.

According to the National Institute of Standards and Technology (NIST) definition of cloud computing [6], key characteristics such as on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service make cloud platforms ideally suited for financial applications. These features ensure that systems can respond dynamically to changes in demand, such as an increase in the number of active users or data transactions, without significant downtime or manual intervention.

Scalability is a core advantage of using cloud infrastructure. Financial applications, including pension management systems, must handle varying workloads as users interact with their accounts, generate reports, or perform data-heavy operations like downloading pension statements. Hosting backend services and databases on a cloud environment ensures the ability to scale horizontally or vertically based on system needs, thus avoiding performance bottlenecks that could degrade user experience [23].

Security is another critical consideration for financial applications in the cloud. Modern cloud platforms offer built-in security features such as encryption at rest and in transit, identity and access management controls, automated backups, and regular compliance auditing [22]. Firebase, for instance, provides secure data storage options and strong access control mechanisms that align with industry standards for financial data protection. Although the deployment for this project is within an academic context, the system architecture adheres to cloud security principles essential for handling real pension data.

High availability and disaster recovery capabilities are additional benefits provided by cloud hosting. Cloud platforms like Firebase and Amazon Web Services (AWS) ensure redundancy and resilience by distributing data across multiple data centres [23]. By using Firebase's distributed infrastructure, the system minimizes the risk of single points of failure and ensures that user access to pension data remains reliable, even during localized service disruptions.

Incorporating cloud computing into the pension management system not only enhances scalability, security, and availability but also aligns with industry trends where major pension providers and financial institutions are increasingly migrating to secure cloud environments [27]. By simulating these real-world practices, the project mirrors current technological advancements in the financial services sector, ensuring the developed solution remains relevant and adaptable for future expansion.

3.4 User Experience (UX) for Digital Financial Platforms

User Experience (UX) is a critical factor in the success of digital financial applications, including pension management systems. Financial services often involve complex data and sensitive user interactions, making it essential to design platforms that are intuitive, accessible, and responsive. Poor usability can result in user frustration, decreased trust, and disengagement, especially among demographics less familiar with digital technologies. Therefore, creating a user-friendly interface is a priority for building a successful pension management tool.

According to Nielsen's usability engineering principles [8], financial platforms must prioritize clarity, simplicity, and ease of navigation. Complex financial data, such as pension contributions and fund performance, must be presented in a way that users can quickly understand without requiring specialized financial knowledge. This project adopts these best practices by using clear dashboard layouts, minimalistic form designs, and concise, jargon-free labelling of key pension metrics.

Accessibility is another important aspect of UX in financial services. The UK Government Digital Service (GDS) guidelines [9] emphasize that digital services must be inclusive, accommodating users with a wide range of physical abilities and technical skills. This project incorporates responsive design principles through Angular and Bootstrap to ensure that the pension management platform is accessible on desktops, tablets, and smartphones. Font sizes, colour contrasts, and button placements are optimized for readability and ease of interaction, making the system more usable for older adults who form a significant portion of the pension-holding population.

Visual data representation is a proven strategy to enhance user comprehension in financial systems. Studies show that graphical elements such as charts and graphs improve users' ability to interpret complex financial information quickly [27]. In this project, Chart.js is used to generate dynamic visualizations of pension fund growth and contribution history. Interactive graphs provide users with insights into their financial progress without overwhelming them with raw numerical data.

Furthermore, user-centred design processes, such as iterative testing and feedback collection, are essential for continuously improving digital financial platforms. Although the scope of this academic project limits full-scale usability testing, the design decisions are informed by established UX research and best practices [8], [9], ensuring a strong foundation for intuitive user interaction.

By embedding strong UX principles into the design of the pension management system, the project aims to improve user trust, engagement, and satisfaction — key factors that are crucial not only for the success of financial applications but also for encouraging proactive retirement planning among users.

3.5 Emerging Technologies: AI, Automation, and Gamification

Emerging technologies such as artificial intelligence (AI), automation, and gamification are reshaping the landscape of digital financial services, offering innovative ways to enhance user engagement, streamline operations, and promote financial literacy. While the core scope of this project focuses on building a secure and user-friendly pension management platform, the potential integration of these emerging technologies provides opportunities for future system enhancements.

AI-powered chatbots have gained popularity in financial platforms for their ability to provide real-time assistance, answer user queries, and offer personalized guidance. Research by Johnson [10] and Patel and Wang [11] demonstrates that AI-driven financial advisors can significantly improve user engagement by making complex pension concepts easier to understand. Integrating a chatbot into the pension management system could allow users to receive instant support for questions related to contribution tracking, fund performance, or general pension planning, thereby enhancing user experience and reducing administrative support requirements.

Automation technologies also hold significant promise for improving financial platforms. In the context of pension management, automated processes can assist with tasks such as contribution calculations based on salary changes, fund rebalancing, and annual statement generation. For this project, basic automation has been incorporated through the automatic calculation of employer and employee contributions based on preset percentage rules. Expanding automation capabilities could further streamline operations and ensure accuracy in managing pension records, in line with emerging best practices in digital financial services [27].

Gamification is another emerging trend aimed at improving user engagement and financial literacy. Studies indicate that incorporating game-like elements, such as badges, progress tracking, and rewards, can encourage users to take more active roles in managing their financial futures [28]. In a pension management context, gamification could be used to motivate users to increase their contributions, review their pension status regularly, or complete educational modules about retirement planning. Although gamification is not implemented in the current version of the system, it presents a valuable direction for future development, particularly in engaging younger users with long-term financial planning.

Overall, emerging technologies like AI, automation, and gamification present transformative opportunities for enhancing digital pension management platforms. While these features are outside the immediate scope of the initial implementation, the system architecture has been designed to be modular and extensible, allowing for the integration of such technologies in future phases. By considering these trends, the project aligns itself with the ongoing evolution of the financial services industry and anticipates user expectations for smarter, more interactive, and more personalized digital experiences.

3.6 Comparison Between the Proposed System and Existing Systems

Existing pension management systems, such as those provided by “NEST” and “The People’s Pension”, offer robust digital platforms that allow users to view contributions, update personal details, and download annual pension statements. These systems are designed for large-scale operation, supporting millions of users with strict adherence to regulatory compliance, including GDPR, FCA, and TPR guidelines [4], [5]. However, despite their maturity, many existing platforms face criticism for lacking modern user interface design, real-time interactivity, and proactive engagement features [9], [19]. The proposed pension management system in this project addresses these gaps by emphasizing a highly intuitive, responsive user experience based on modern web technologies such as Angular and Bootstrap, supported by dynamic fund visualizations using Chart.js. Unlike traditional systems that often provide static information, the developed platform offers real-time contribution tracking, immediate access to downloadable monthly statements, and simulated fund performance insights. Furthermore, while existing systems mainly focus on compliance and basic functionality, this project lays the foundation for future enhancements through modular system design, enabling the addition of AI-powered chatbots [10], [11], multi-language support, automation [27], and gamification features [28] to drive user engagement. Thus, while existing platforms remain comprehensive in terms of compliance and scale, the proposed system introduces innovations in user experience, personalization, and extensibility, providing a forward-looking model for next-generation pension management applications.

4. System Design

4.1 System Architecture

The pension management platform is designed using a three-tier client-server architecture, which separates the system into distinct layers to enhance scalability, maintainability, and security. This architecture model allows for a clear division of responsibilities between the user interface, application logic, and data storage, facilitating modular development and future extensibility.

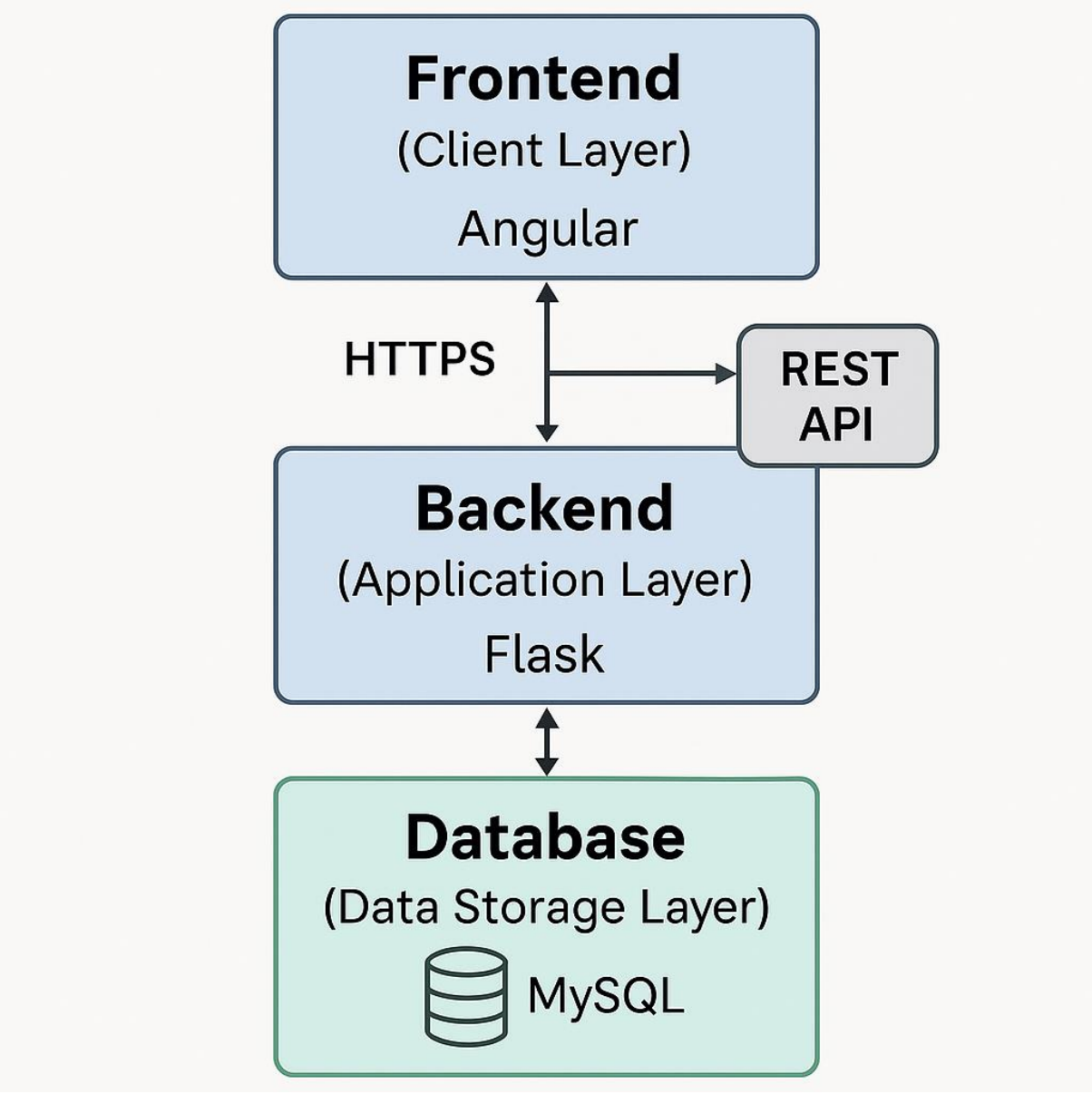


Fig 1. System Architecture

1. Frontend (Client Layer)

The frontend is developed using Angular, a popular framework for building responsive and scalable single-page applications. The client-side interface is responsible for rendering content, managing user interactions, and securely communicating with backend services via RESTful APIs.

Key functionalities include:

- Secure login forms with multi-factor authentication (MFA).
- Dynamic dashboards displaying monthly pension statements in tabular and graphical (Chart.js) formats.
- User profile and beneficiary management interfaces for personal information updates.
- Document upload and file viewing via protected routes.
- Support for multi-language UI (English, Hindi, Polish, French, German) via ngx-translate.
- Chatbot integration using Voiceflow, enabling AI-powered guidance for users.
- Support for downloading official pension statements in PDF format.

2. Backend (Application Layer)

The backend is built using **Flask**, a lightweight Python web framework. It provides RESTful API endpoints and enforces robust authentication, authorization, and data validation mechanisms.

Responsibilities include:

- Handling user login, session control, and issuing secure JWTs.
- Enforcing Role-Based Access Control (RBAC) for employees, employers, and admins.
- Processing uploaded pension contribution data and calculating values dynamically.
- Managing monthly pension statement generation and secure PDF creation.
- Secure file handling for document upload and download modules.
- Structured into modular route files: auth_routes.py, admin_routes.py, employer_routes.py, and others for clarity and maintainability.
- Includes basic error handling, JWT expiration checks, and validation for user sessions and route access.

3. Database (Data Storage Layer)

The system uses MySQL, a robust relational database management system ideal for handling structured pension data. Interaction with the database is managed through Flask-SQLAlchemy ORM to ensure efficient and secure querying.

Stored data includes:

- User accounts, hashed passwords, and role-specific metadata.
- Monthly pension statements, contribution breakdowns, and fund performance logs.
- Uploaded document metadata (file path, type, timestamps).
- Beneficiary details and allocation percentages.

Security & Integrity:

- Sensitive fields are encrypted where appropriate.
- The database is not exposed directly; all operations go through secure backend endpoints.
- Referential integrity is enforced through foreign key constraints and ORM relationships.

4. Communication Flow

- The user interacts with the Angular frontend over a secure HTTPS connection.
- API calls from the frontend include JWTs for session validation.
- The Flask backend processes requests, verifies tokens, performs role-based checks, and interacts with the MySQL database as needed.
- Responses (e.g., pension data, PDF download links, translated UI labels) are sent back securely.
- Files and documents are delivered via protected endpoints with access control in place.

This architecture ensures modular development, secure operation, and scalable deployment. The layered structure supports core pension functionality while also allowing future integration of features such as AI-driven financial forecasting or additional language packs. The implemented chatbot and multi-language interface already demonstrate extensibility and commitment to inclusive, accessible financial technology.

4.2 Pension Provider Simulation

The pension management platform developed in this simulates a UK Workplace Pension Scheme, modelled after leading real-world providers such as “NEST” and “The People’s Pension”. These providers operate under the UK government’s auto-enrolment legislation, requiring both employers and employees to contribute a percentage of salary toward a retirement savings account.

The system replicates key aspects of modern pension schemes, including monthly contribution processing, fund performance tracking, statement generation, and secure role-based access. Through these features, it delivers a practical and realistic simulation of how workplace pension platforms operate in the UK financial ecosystem.

Core Elements of the Simulation

1. Monthly Contributions

Employers and employees contribute a predefined percentage of the employee's salary to the pension fund each month:

- **Employer contribution:** typically 3% of qualifying earnings
- **Employee contribution:** typically 5%
- **Total:** 8% monthly contribution

These values are calculated dynamically within the system based on user-input salary data and stored securely in the MySQL database. This mirrors the automatic contribution calculations performed by real-world payroll systems.

2. Pension Statements

Each month, upon employer contribution upload, a digital pension statement is automatically generated for every employee, summarizing:

- Employer and employee contributions
- Total monthly contribution
- Fund performance (growth rate)
- Updated pension pot value

Statements are:

- Viewable in-tabular format via the dashboard
- Downloadable in PDF format for secure recordkeeping
- Linked with real-time notifications that inform employees of new statement availability

This behaviour mirrors platforms like "NEST", which allow members to view/download statements through secure portals.

3. Fund Performance Tracking

The system applies a fixed annualized growth rate to simulate fund performance, similar to diversified pension fund modelling.

Key features:

- Dynamic fund growth charts using Chart.js
- Month-over-month visualization of pot value growth
- User-friendly analytics panel embedded in the dashboard

4. Role-Based Access and Security

The platform enforces multiple layers of protection to simulate institutional-grade data privacy:

- Multi-factor authentication (MFA) for all users
- Role-Based Access Control (RBAC) separating employee, employer, and admin permissions
- JWT-based session management and secure API access
- Encrypted communication via HTTPS/TLS

Users can only view data relevant to their identity and role, emulating the strict access controls used by regulated pension institutions.

5. Regulatory Alignment
The system simulates compliance with key UK regulatory and governance frameworks:

- TPR Compliance – Automatic monthly contributions and structured pension reporting reflect guidelines set by The Pensions Regulator.
- GDPR Compliance – Personal data is encrypted, access-controlled, and editable/deletable on request.
- FCA-Inspired Transparency – The interface and statements ensure clarity, accuracy, and accessibility, aligned with standards of the Financial Conduct Authority (FCA).

6. Employer and Administrator Workflows
The system enables employers to:

- View employees under their domain
- Track their employees' pension records

Administrators simulate the platform operator's role by:

- Adding/removing employers
- Viewing platform-wide analytics
- Generating or deleting statements system-wide

7. Beneficiary Integration
Employees can:

- Add and update beneficiaries
- Assign allocation percentages
- View/manage these records as part of their profile

This feature enhances the realism of the platform and reflects actual pension scheme structures where beneficiaries must be recorded for legal compliance. This simulation replicates both the technical and procedural realities of workplace pension systems used in the UK. It delivers an authentic experience for users, employers, and admins while ensuring security, compliance, and engagement all critical benchmarks for digital financial systems.

4.3 Statement Structure

A core feature of the pension management platform is the generation and structured presentation of monthly pension statements. These documents allow employees to track their retirement savings, monitor contributions, and assess fund performance all while ensuring clarity, transparency, and alignment with real UK workplace pension standards.

Each statement is auto-generated following the upload of monthly contributions by employers and can be viewed on the user dashboard or downloaded securely in PDF format.

1. Basic Information
Each statement begins with core identifying details:

- **Statement ID**: A unique identifier (e.g., PS2025-05-001)
- **Statement Period**: Coverage dates (e.g., 01 May 2025 – 31 May 2025)
- **Statement Date**: Date of issuance (e.g., 31 May 2025)

- **Full Name**: Name of the employee
- **NI Number**: National Insurance number (partially masked)
- **Pension Account Number**: Unique account reference (e.g., PB-0001)
- **Employer Name**: The associated organization (e.g., Acme Corp)

2. Contribution Details
The statement provides a transparent view of monthly contributions:

- **Employer Contribution**: e.g., £3,000.00
- **Employee Contribution**: e.g., £5,000.00
- **Total Contribution**: Combined monthly contribution (e.g., £8,000.00)
- **Contribution Date**: Date of processing
- **Contribution % Breakdown**: Typically Employer: 3%, Employee: 5%
- **Basis of Calculation**: Derived from recorded annual salary and stored securely in the backend database

3. Fund Information
Each statement includes details about the investment and pot performance:

- **Pension Fund Name**: e.g., "Medium Risk Fund"
- **Fund Performance**: Simulated growth for the period (e.g., +2.50%)
- **Risk Profile**: Low, Medium, or High (based on employee selection)
- **Current Pot Value**: Cumulative pension savings after growth (e.g., £8,000.00)

4. Statement Access Options
Statements are made available through the employee dashboard:

- **Dashboard Table View**: All monthly statements are shown in a sortable table
- **PDF Download**: Each entry includes a secure link to download the statement as a PDF
- **Real-Time Notifications**: Users receive alerts when a new statement is generated

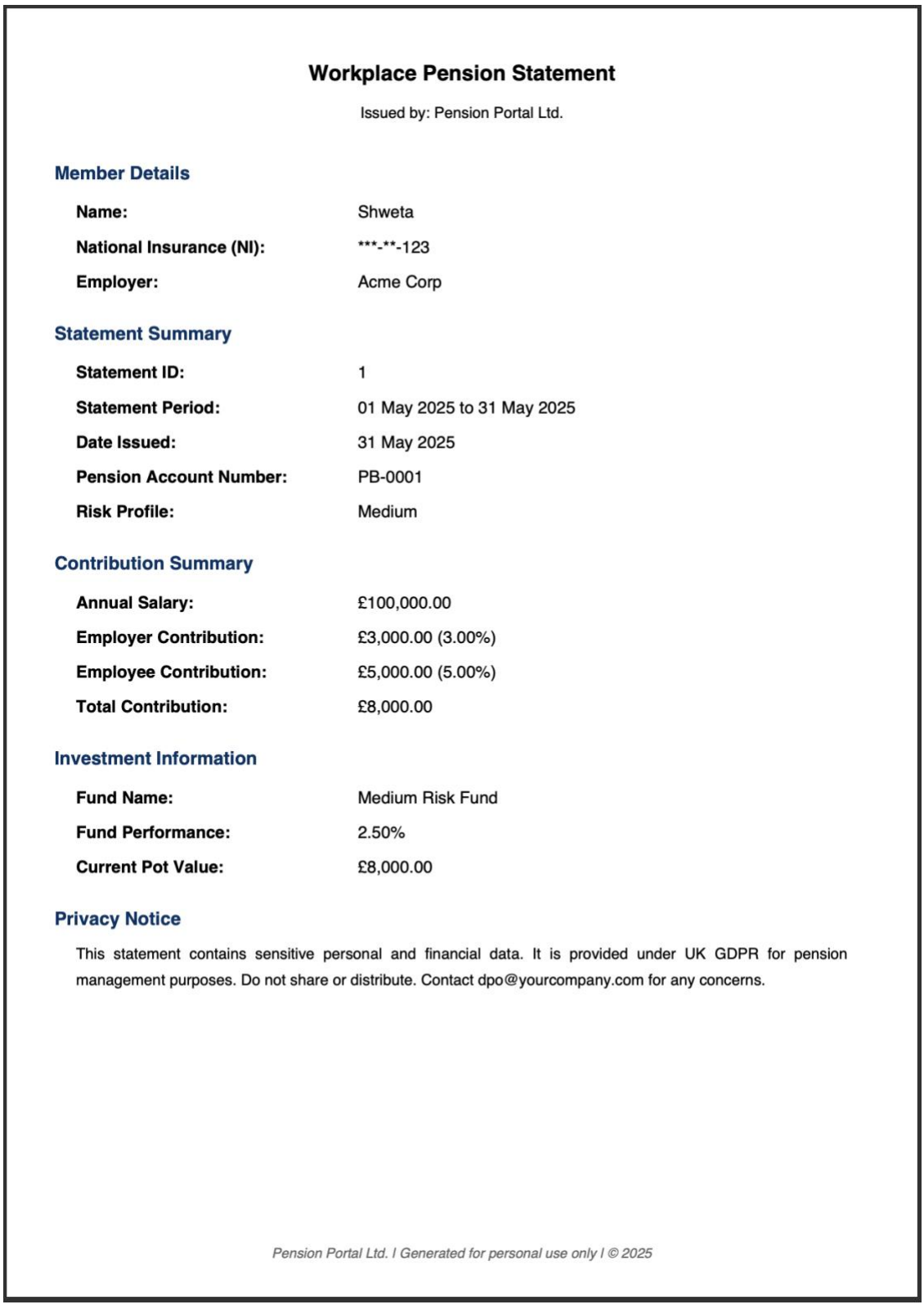
5. Security and Confidentiality Notice
Each PDF and dashboard view includes the following message:
"This statement is confidential and intended for authorized users only. Please contact your pension administrator for any queries or assistance. This document is for personal use and should not be distributed without consent."

6. Sample Dashboard Table Layout

Date	Employee Contribution	Employer Contribution	Total	Fund Name	Actions
May 2025	£5,000.00	£3,000.00	£8,000.00	Medium Risk	[View] [PDF]
Apr 2025	£4,800.00	£2,880.00	£7,680.00	Medium Risk	[View] [PDF]

This detailed and compliant format ensures employees are always informed about their pension performance and contributions while aligning with real-world standards for digital pension documentation.

7. Sample PDF Layout



Workplace Pension Statement

Issued by: Pension Portal Ltd.

Member Details

Name:	Shweta
National Insurance (NI):	***-**-123
Employer:	Acme Corp

Statement Summary

Statement ID:	1
Statement Period:	01 May 2025 to 31 May 2025
Date Issued:	31 May 2025
Pension Account Number:	PB-0001
Risk Profile:	Medium

Contribution Summary

Annual Salary:	£100,000.00
Employer Contribution:	£3,000.00 (3.00%)
Employee Contribution:	£5,000.00 (5.00%)
Total Contribution:	£8,000.00

Investment Information

Fund Name:	Medium Risk Fund
Fund Performance:	2.50%
Current Pot Value:	£8,000.00

Privacy Notice

This statement contains sensitive personal and financial data. It is provided under UK GDPR for pension management purposes. Do not share or distribute. Contact dpo@yourcompany.com for any concerns.

Pension Portal Ltd. | Generated for personal use only | © 2025

Fig 2: PDF Layout

4.4 Use Case Diagram

The use case diagram provides a high-level representation of the interactions between the system and its various users. It identifies the primary actors involved in the pension management system Employee, Employer, Admin, and AI Chatbot—and illustrates the specific functions (use cases) each actor is permitted to perform. This modelling helps define the system's scope and the responsibilities of each user role in alignment with real-world pension management workflows.

Primary Actors and Use Cases

Employee
Employees represent the end users of the pension platform. Their primary focus is on viewing and managing their pension-related information.

- Login / Logout
- Reset Password
- View/Edit Personal Profile
- View Pension Statements
- Download Pension Statements (PDF)
- Upload / View / Download Documents
- Manage Beneficiaries
- View Fund Growth (Charts)
- Receive Notifications (e.g., new statement generated)
- Interact with AI Chatbot
- Change Language (Multilingual UI)

Employer
Employers are responsible for managing their employees and contributing to their pension records.

- Login / Logout
- Reset Password
- View/Add Employees
- View Employee Statements
- View/Update Own Profile
- Interact with AI Chatbot
- Change Language (Multilingual UI)

Admin
Admin manage the overall system, including employer onboarding and high-level pension statement operations.

- Login / Logout
- Add/Delete Employers
- View All Employers and Employees
- Generate Statements
- Delete Statements
- Interact with AI Chatbot
- Change Language (Multilingual UI)

AI Chatbot (System Actor)

The integrated chatbot provides users with quick assistance and educational responses.

- Answer questions related to workplace pensions in the UK

The following diagram illustrates the interactions between each system actor and their associated use cases. It visually captures the system boundaries and role-specific operations within the pension management platform:

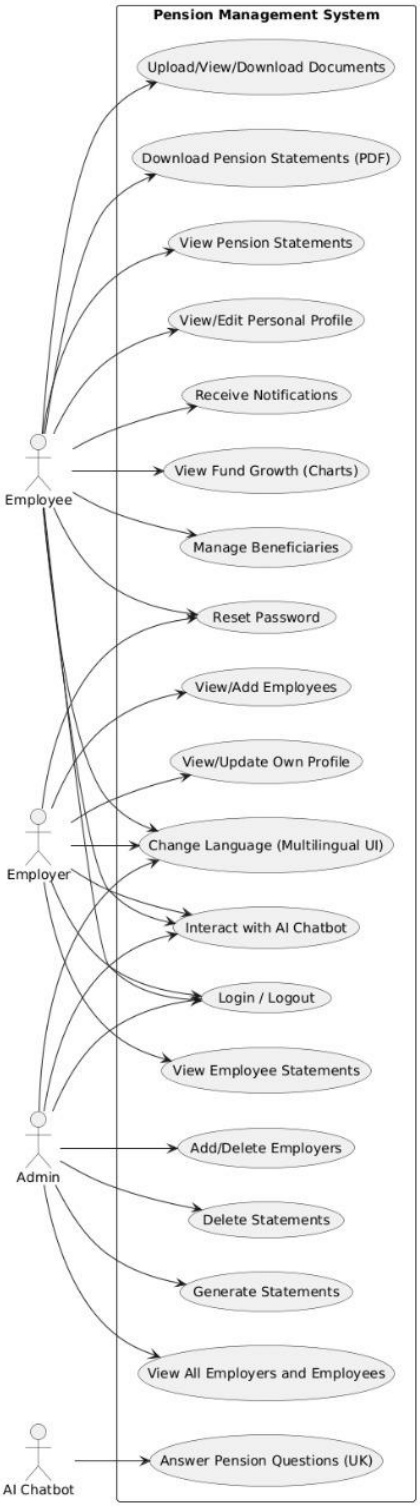


Fig 3: Use Case Diagram

This diagram demonstrates a clearly modular and role-based access structure. Each actor interacts only with functions relevant to their role. For instance, while employees can manage their personal and beneficiary details, only employers can upload contribution data, and only admins have rights to delete or generate system-wide statements. The inclusion of a multilingual interface and chatbot enhances usability and accessibility, aligning the platform with modern user-centric design expectations and accessibility standards in UK financial services.

4.5 Database Design

The pension management platform is supported by a structured and normalized relational database, implemented using MySQL and accessed via Flask-SQLAlchemy ORM. The database schema, named pension_db, is designed to support core system functionalities including secure authentication, role-based access, pension statement tracking, document storage, and beneficiary management.

Core Tables and Structures

1. **users**
   Central table for all user roles (Employees, Employers, Admins).
   - Fields: id, name, email, password (hashed), role, mfa_secret, employer_id (FK), current_salary, ni_number, investment_type, etc.
   - Supports MFA, JWT, and RBAC.
   - Linked to pension_statements, beneficiaries, and documents.
2. **admins**
   Holds admin-specific login and audit details.
   - Fields: id, name, email, password, role, created_at, is_active, last_login
   - Logical mapping to users table via shared email.
3. **employers**
   Contains information about registered employers.
   - Fields: id, name, email, contact_number, address, password, mfa_secret
4. **pension_statements**
   Monthly records of contributions and fund performance.
   - Fields: id, user_id (FK), salary, employee_contribution, employer_contribu tion, total_contribution, fund_performance, pot_value, statement_date, etc.
5. **beneficiaries**
   Stores pension nominee details.
   - Fields: id, user_id (FK), full_name, relationship, percentage
6. **documents**
   Tracks uploaded pension-related files.
   - Fields: id, user_id (FK), filename, filepath, document_type, uploaded_at
7. **alembic_version** *(System Table)*
   Automatically created by Alembic to manage database migrations.
   - Not directly used in application logic.

Table Relationship Mapping (Foreign Keys)

Source Table	FK Field	Target Table	PK Reference	Relationship
users	employer_id	employers	id	Many-to-One
beneficiaries	user_id	users	id	Many-to-One
documents	user_id	users	id	Many-to-One
pension_statements	user_id	users	id	Many-to-One
admins	email	users	email	Logical Match (1:1)

Entity Relationship Summary

1. **Admin → User**
    - Admins are managed both in admins and users tables. Logical linkage is maintained via the email field, enabling flexible role handling.
2. **User → Employer**
    - Each user may belong to one employer (users.employer_id → employers.id). One employer can have multiple users.
3. **User → Pension Statements / Documents / Beneficiaries**
    - Each user can have multiple pension statements, documents, and beneficiaries. One-to-many relationships are enforced with foreign keys.

How This Supports System Design

- **Modularity**: Each table handles a specific domain (users, finance, documents).
- **Referential Integrity**: All foreign keys enforce consistency across linked data.
- **Security & RBAC**: Role-based access logic is implemented via the role column in users.
- **Scalability**: New features (audit logs, fund types, advisor chat logs) can be integrated easily without altering existing relationships.

Data Security and Privacy Measures

- Passwords are hashed with Flask-Bcrypt.
- MFA enabled via mfa_secret.
- Session validation through JWTs with expiry and token blacklisting.
- GDPR principles observed via data update/delete routes and role-based access filters.

Normalization and Relationship Model

- **One-to-Many**:
    - A user can have many pension statements, documents, and beneficiaries.
- **Many-to-One**:
    - Multiple users can belong to one employer.
- **No redundancy**: Tables are normalized up to at least 3NF

Automation and Backup

- Monthly statements are generated via scheduled Flask tasks.
- Daily MySQL backups are automated to a secure storage location.

- The schema supports analytics queries (e.g., total contributions per employer) and logging extensions.

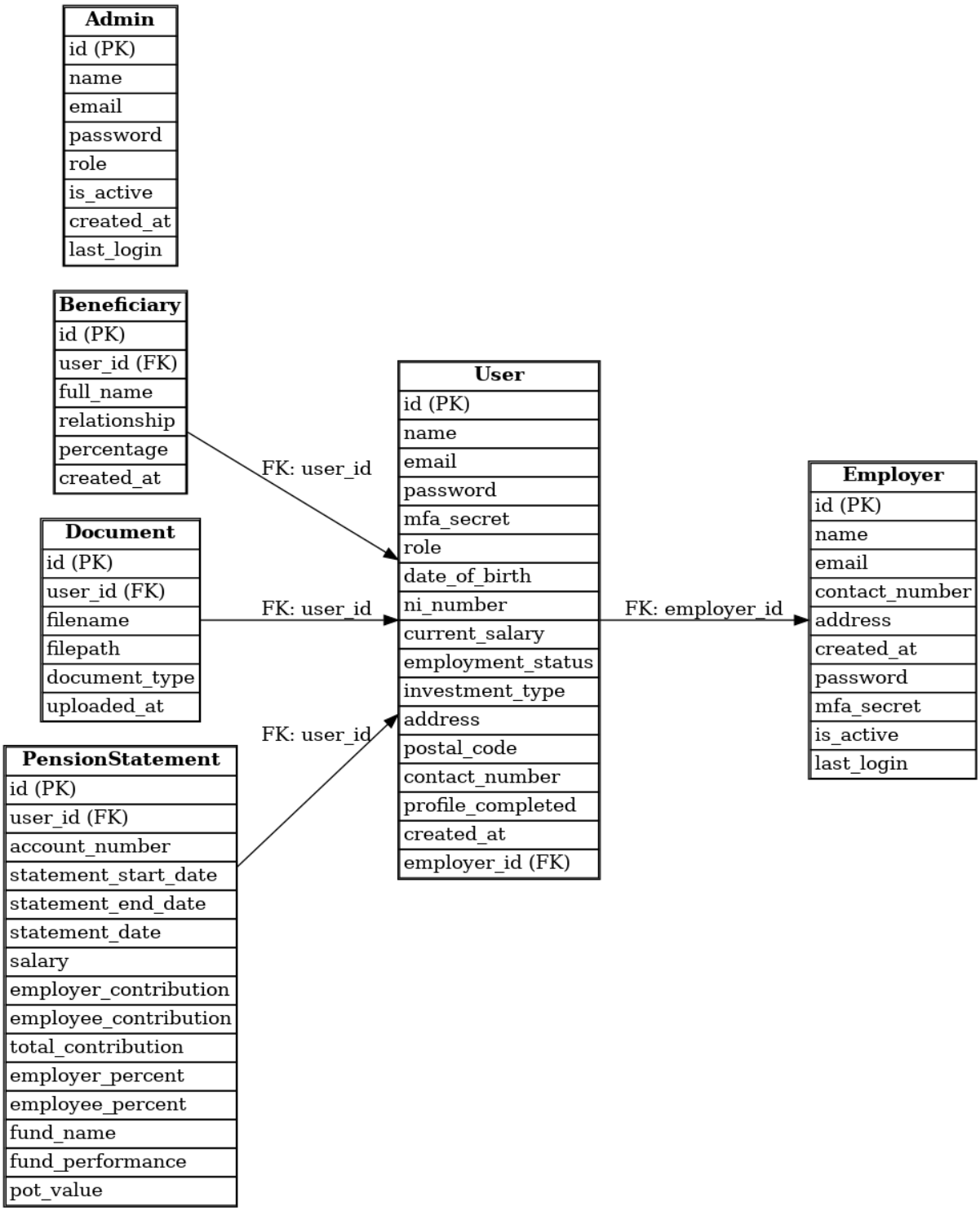


Fig 4: ER Diagram

4.6 Technology Stack Justification

The selection of the technology stack for this pension management platform was guided by key considerations such as scalability, security, regulatory compliance, modular architecture, and developer productivity. The chosen tools and frameworks enable rapid development while ensuring that the system adheres to high standards in user experience, performance, and data protection.

1. Frontend: Angular

- **Justification**:
  - Angular is a mature, enterprise-grade framework ideal for scalable single-page applications (SPAs).
  - Built-in support for component-based architecture promotes modularity, reusability, and clean code organization.
  - Native support for routing, form validation, and HTTP services streamlines frontend-backend communication.
  - Integrates seamlessly with third-party libraries like ngx-translate for i18n and Chart.js for data visualization.
  - Excellent support for responsive design using Bootstrap and Font Awesome for styling and iconography.

2. Backend: Flask (Python)

- **Justification**:
  - Flask is a lightweight and flexible micro-framework suitable for building RESTful APIs quickly.
  - Minimal overhead makes it ideal for rapid prototyping and customization of security/authentication workflows.
  - Compatible with essential backend tools like JWT (Flask-JWT-Extended), Flask-Bcrypt, and Flask-Mail.
  - Well-suited for modular API routing (e.g., admin_routes.py, user_routes.py) and RBAC enforcement.

3. Database: MySQL

- **Justification**:
  - MySQL is a widely adopted relational database system that offers robust security, data integrity, and ACID compliance.
  - Supports normalized schema design to maintain clear relationships between users, employers, statements, and beneficiaries.
  - Well-integrated with Flask-SQLAlchemy, an ORM that provides abstraction, ease of use, and support for migrations (via Flask-Migrate).

4. Authentication & Security:

- **JWT (JSON Web Tokens)**: Used for stateless, secure session management, ensuring scalability and protection against session hijacking.
- **Flask-Bcrypt**: For strong password hashing using salted hashes.
- **MFA (TOTP)**: Time-based One-Time Passwords implemented for high-privilege roles to strengthen login security.
- **HTTPS/TLS**: Enforced in production to ensure encrypted API communication and GDPR compliance.

5. PDF Generation: (Python)

- **Justification**:
  - Enables custom PDF generation for monthly pension statements with programmatic control over layout and content.
  - Suitable for secure, non-editable document delivery aligned with financial reporting standards.

6. Internationalization (i18n): ngx-translate

- **Justification**:
  - Simplifies language switching and supports multilingual UI without altering component logic.
  - Allows centralized JSON-based translation management (e.g., en.json, hin.json, etc.).

7. Chatbot: Voiceflow

- **Justification**:
  - Allows creation of intuitive conversational AI flows without needing complex backend AI training.
  - Enhances user support, especially for low-digital-literacy users, by answering pension-related queries in real time.
  - Easily embedded via a script in the Angular frontend (index.html) with secure configuration.

This carefully curated technology stack balances simplicity and extensibility, enabling rapid development without compromising performance or security. The use of Angular and Flask ensures maintainability and separation of concerns, while MySQL and supporting libraries offer data reliability and protection. Together, these technologies provide a strong foundation for a secure, scalable, and user-centric pension management platform.

5. Implementation

5.1 Frontend Development

The frontend of the pension management platform is developed using Angular, a modern, TypeScript-based framework that facilitates scalable, modular, and maintainable single-page applications (SPAs). The user interface (UI) is designed to deliver an intuitive, secure, and accessible experience for three distinct user roles: Admin, Employer, and Employee. Built on a component-driven architecture, the frontend ensures clear separation of concerns, seamless data binding, and maintainability. It integrates with backend services through Angular's HttpClient for consuming REST APIs securely and efficiently.

Design and Responsiveness
The UI is fully responsive, achieved through Bootstrap 5, ensuring usability across desktops, tablets, and mobile devices. UI icons and feedback mechanisms are enhanced using FontAwesome to improve visual clarity and usability.

Core Component Architecture
Angular's modular system enables development of dedicated components for each functional area:

Authentication & Security
- login.component: Implements login with MFA, email, and password.
- request-reset.component / reset-password.component: Handles secure password recovery and reset flows.
- Route guards (AuthGuard) enforce role-based navigation and secure route access.

Role-Specific Dashboards
- admin-dashboard.component: Displays overall system statistics, user management tools, and employer controls.
- employer-dashboard.component: Allows employers to manage employee accounts and initiate password reset requests.
- dashboard.component: Provides employees with personalized dashboards showing pension summaries and actions.

Pension Statement Management
- user-statements.component: Enables employees to view and download monthly pension statements.
- analytics.component: Integrates Chart.js to visualize fund growth and contribution history.
- add-user.component: Allows employers to register new employees under their organization.

Profile & Beneficiary Management
- profile.component: Lets users update personal details including NI number, salary, and contact info.
- beneficiary.component: Employees can add and manage pension beneficiaries with allocation percentages.

Document Management

- upload-document.component: Users can securely upload and manage pension-related documents.
- user-documents.component: Displays uploaded files with metadata and download options.

Internationalization (i18n)

- Implemented using ngx-translate, enabling dynamic switching between languages.
- Fully supports English (en.json) , Hindi (hin.json), Polish (pl.json), French (fr.json), German (de.json).
- Key static pages (Home, About, FAQ) and dashboard labels are integrated with translation services.

AI Chatbot Integration

- A Voiceflow-based floating chatbot is embedded via JavaScript in index.html.
- Offers real-time responses to pension-related FAQs and helps with navigation assistance.
- Enhances accessibility, particularly for users with limited digital literacy.

Form Validation & UX Enhancements

- Uses Angular reactive forms and template-driven forms for robust validation.
- Features like input masking, validation messages, and dialog confirmations improve usability.
- Consistent styling across modals, buttons, alerts, and layouts enhances overall user experience.

Frontend-Side Security

- JWT tokens are securely stored in localStorage and attached to HTTP headers for all authenticated API requests.
- MFA is enforced during admin and employer login flows.
- Route-level protection ensures users cannot navigate to unauthorized areas based on their role.

Routing and API Integration

- API endpoints from the Flask backend are integrated via Angular services.
- Routes are managed via app.routes.ts and configured in app.config.ts, segregating access by user role.
- Navigation guards ensure that users remain within their authorized scope.

Extensibility and Future Enhancements

- The frontend is built with scalability in mind, allowing seamless integration of future features such as:
    - Gamification elements
    - Multilingual support beyond Hindi , Polish, French, German and English
    - Voice command access
    - AI-driven retirement projections

5.2 Backend Development

The backend of the pension management system is developed using **Flask**, a lightweight and modular Python web framework. It acts as the core engine that handles all business logic, processes data operations, and enforces security across different user roles. The backend exposes RESTful APIs consumed by the Angular frontend, facilitating secure, role-based interactions between users and the system.

Technology Stack

Layer	Technology
Web Framework	Flask (Python)
ORM	Flask-SQLAlchemy
Authentication	JWT (Flask-JWT-Extended) + Bcrypt
Database	MySQL
PDF Generation	pdf_generator.py
Scheduler	APScheduler
Email Services	Flask-Mail (SMTP)

Authentication & Security

The backend employs a robust security framework to ensure protected access and data integrity:

- **JWT Authentication**: Secure, stateless login using JSON Web Tokens for all protected API requests.
- **Multi-Factor Authentication (MFA)**: Employers and admins must provide a valid 6-digit TOTP during login.
- **Password Hashing**: All user passwords are stored using Flask-Bcrypt with strong salting algorithms.
- **Token Blacklisting**: Implements a blacklist system to invalidate tokens on logout or session expiry.
- **CORS Policy**: Configured to allow secure cross-origin communication between the frontend and backend.

Modular API Structure

The backend routes are modularized by user role for scalability and maintainability:

- **auth_routes.py** – Handles login, logout, registration, and password resets.
- **admin_routes.py** – Manages admin-specific tasks: user/employer creation, deletion, and analytics.
- **employer_routes.py** – Enables employers to manage employee accounts and generate contribution data.
- **user_routes.py** – Allows employees to manage their profile, view documents/statements, and update beneficiaries.

Each module includes:

- Input validation
- Role-based access checks

- Consistent response formatting
- Error handling with HTTP status codes and JSON feedback

Pension Statement Generation

Pension contributions and statements are dynamically calculated based on salary and stored securely:

- **Employer Contribution**: 3%
- **Employee Contribution**: 5%
- **Total**: 8% of gross salary per month
- **Fund Simulation**: Monthly simulated growth rate (e.g., 1%) applied to the pension pot
- **PDF Rendering**: Statements generated via ReportLab, stored and linked for frontend download

Scheduled Tasks

Using APScheduler, the backend automates time-based tasks such as:

- Monthly pension statement generation
- Triggering statement notification flags
- Optional: Periodic backups or admin alerts (for future scalability)

Document Management

Users can upload documents (PDF, CSV) via secure endpoints:

- Metadata such as filename, type, upload time, and user_id is stored in the database.
- Files are stored securely in a restricted server directory.
- Download and access is protected by JWT-based user session verification.

Password Recovery & Email Integration

Password recovery is secured via tokenized email flow:

- A token is generated upon password reset request and emailed to the user via **Flask-Mail**.
- The user can reset their password using a protected /reset-password route.
- All token links expire after a configurable time window for enhanced security.

Security Practices

- HTTPS is enforced via SSL certificates in production deployments.
- All critical endpoints are protected using JWTs and role-check decorators.
- Access logs and audit trails are enabled for sensitive operations like login, updates, and deletions.
- Custom middleware ensures expired or tampered tokens are rejected gracefully.

Testing & Validation

- All API endpoints are tested using **Postman collections**.
- Minimum 7–10 test cases are created per endpoint covering:
    - Success scenarios
    - Invalid/missing fields
    - Authorization failure
    - Token expiration or blacklisting

The backend implementation is designed to ensure high reliability, modularity, and compliance with secure data handling standards. Through its layered architecture, Flask-based routing, JWT security, PDF handling, and scheduler integration, it provides a solid foundation for current functionality and future enhancements like fund switching, AI forecasting, or dynamic employer dashboards.

5.3 Database Implementation

The pension management system utilizes a MySQL relational database to securely store and manage all critical data, including user profiles, employer details, pension statements, beneficiaries, and uploaded documents. The database integrates seamlessly with the Flask backend through Flask-SQLAlchemy, an Object Relational Mapper (ORM) that abstracts raw SQL queries into Python class structures, ensuring efficient and maintainable data access.

Database Configuration
The database schema is named pension_db, and its connection is configured within config.py. Flask manages the session lifecycle and commits transactions using SQLAlchemy's native support.

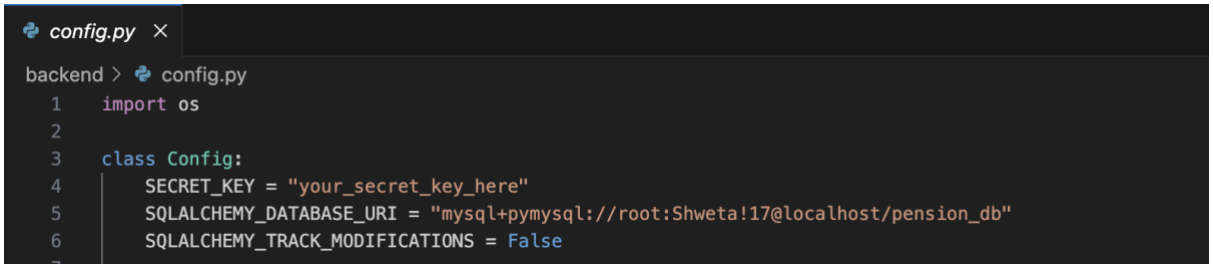


Fig 5: Database Configuration

Model Implementation
Each table is represented by a Python class that extends db.Model. Key models include:

- **User**: Holds core account information including hashed passwords, MFA secret, salary, and employment status.
- **Employer**: Stores organization-specific information such as contact and license details.
- **Admin**: Tracks admin-specific metadata such as login timestamps and role flags.
- **Pension Statement**: Logs monthly contributions, calculated pot value, and simulated fund performance.
- **Beneficiary**: Lists pension nominees with allocation percentages and relationships.
- **Document**: Manages user-uploaded files with type tagging and timestamps.

Each model enforces:

- nullable=False where appropriate
- Indexing for primary and foreign keys
- Default values and data type constraints (e.g., Enum, Numeric)
- Referential integrity using db.ForeignKey() and ORM db.relationship() mappings

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- Default values and data type constraints (e.g., Enum, Numeric)
- Referential integrity using db.ForeignKey() and ORM db.relationship() mappings

Security and Constraints

- **Password Security**: All passwords are stored using Flask-Bcrypt hashing.
- **Role Management**: The role field in the users table governs role-based access (Admin, Employer, Employee).
- **MFA Support**: MFA secrets are stored in mfa_secret column and used for high-privilege user login.
- **Field Validation**: Enum fields such as employment_status and investment_type ensure standardized input.
- **Data Integrity**: Foreign keys prevent orphaned records and maintain data relationships.

Schema Relationships:

Table	Relationship
users	Linked to employers via employer_id
beneficiaries	Linked to users via user_id
pension_statements	Linked to users via user_id
documents	Linked to users via user_id

These mappings support relational queries and enable cascade behaviours, such as fetching all statements for a user or all users under a specific employer.

Migrations and Setup:

- **Initial Table Creation**: During app initialization, db.create_all() generates the required tables.
- **Schema Migrations**: Managed using Flask-Migrate, allowing version-controlled schema updates.
- **Seeding**: Sample data (admins, test users) is populated using custom CLI scripts or Flask shell sessions.

Performance and Maintenance:

- **Index Optimization**: Primary and foreign keys are indexed to accelerate joins and lookups.
- **Data Validation**: Backend APIs validate form inputs before writing to the database to prevent corruption.
- **Referential Integrity**: Cascading delete/update behaviour ensures clean and consistent data structures.

This normalized and modular database implementation provides a robust foundation for secure, scalable, and compliant pension data management. With ORM integration, role-aware relationships, and automated statement generation, the database ensures real-time consistency across modules while adhering to best practices in GDPR compliance and financial application standards.

5.4 Highlighted Code Areas

This section presents six core code segments that demonstrate critical functionality in the pension management system. Each snippet highlights a key technical contribution—ranging from secure authentication and PDF generation to real-time analytics and multilingual UI support. These implementations collectively reflect the system's architectural integrity, security-first approach, and usability focus.

1. JWT-Based Authentication with Multi-Factor Login

File: auth_routes.py
Purpose: Secure login with password verification, MFA token validation, and JWT issuance.

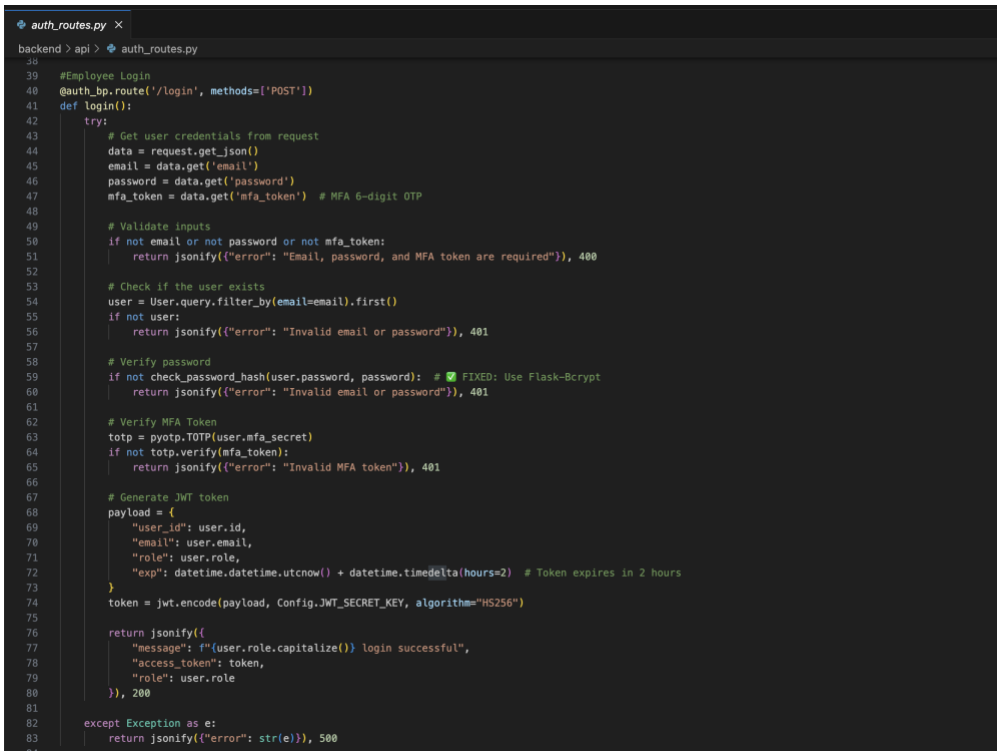


Fig 6: JWT-Based Authentication with Multi-Factor Login

Key Features:

- Secure password validation via Flask-Bcrypt
- TOTP-based multi-factor authentication (for privileged users)
- JWT-based stateless login and access management

2. Pension Statement Generation and Financial Calculations

File: scheduler_tasks.py

Purpose: Automates monthly pension statement creation using dynamic salary-based formulas.

scheduler_tasks.py M

backend > tasks > scheduler_tasks.py

```python
def generate_monthly_statements():
    print("Running Monthly Pension Statement Auto-Generator...")

    today = datetime.date.today()
    first_day = today.replace(day=1)
    last_day = (first_day + datetime.timedelta(days=32)).replace(day=1) - datetime.timedelta(days=1)

    users = User.query.all()

    for user in users:
        if not user.current_salary or not user.date_of_birth:
            continue

        salary = float(user.current_salary)
        employee_percent = 5.0
        employer_percent = 3.0
        employee_contribution = round(salary * (employee_percent / 100), 2)
        employer_contribution = round(salary * (employer_percent / 100), 2)
        total_contribution = employee_contribution + employer_contribution

        # Simulate fund growth by investment type
        performance_map = {"Low": 2.5, "Medium": 5.0, "High": 8.0}
        growth_rate = performance_map.get(user.investment_type or "Medium", 5.0)

        last_pot = db.session.query(func.max(PensionStatement.pot_value)).filter_by(user_id=user.id).scalar() or 0.0
        new_pot = round(last_pot + total_contribution + (last_pot * (growth_rate / 100) / 12), 2)

        statement = PensionStatement(
            user_id=user.id,
            account_number=f"ACCT{user.id:05}",
            statement_date=today,
            statement_start_date=first_day,
            statement_end_date=last_day,
            salary=salary,
            employee_percent=employee_percent,
            employer_percent=employer_percent,
            employee_contribution=employee_contribution,
            employer_contribution=employer_contribution,
            total_contribution=total_contribution,
            fund_name=user.investment_type or "Medium",
            fund_performance=growth_rate,
            pot_value=new_pot
        )

        db.session.add(statement)
        send_statement_notification(user.email, user.name)
```

Fig 7: Pension Statement Generation and Financial Calculations

Key Features:

- Dynamic financial calculations (8% total contribution)
- Pot value growth simulation

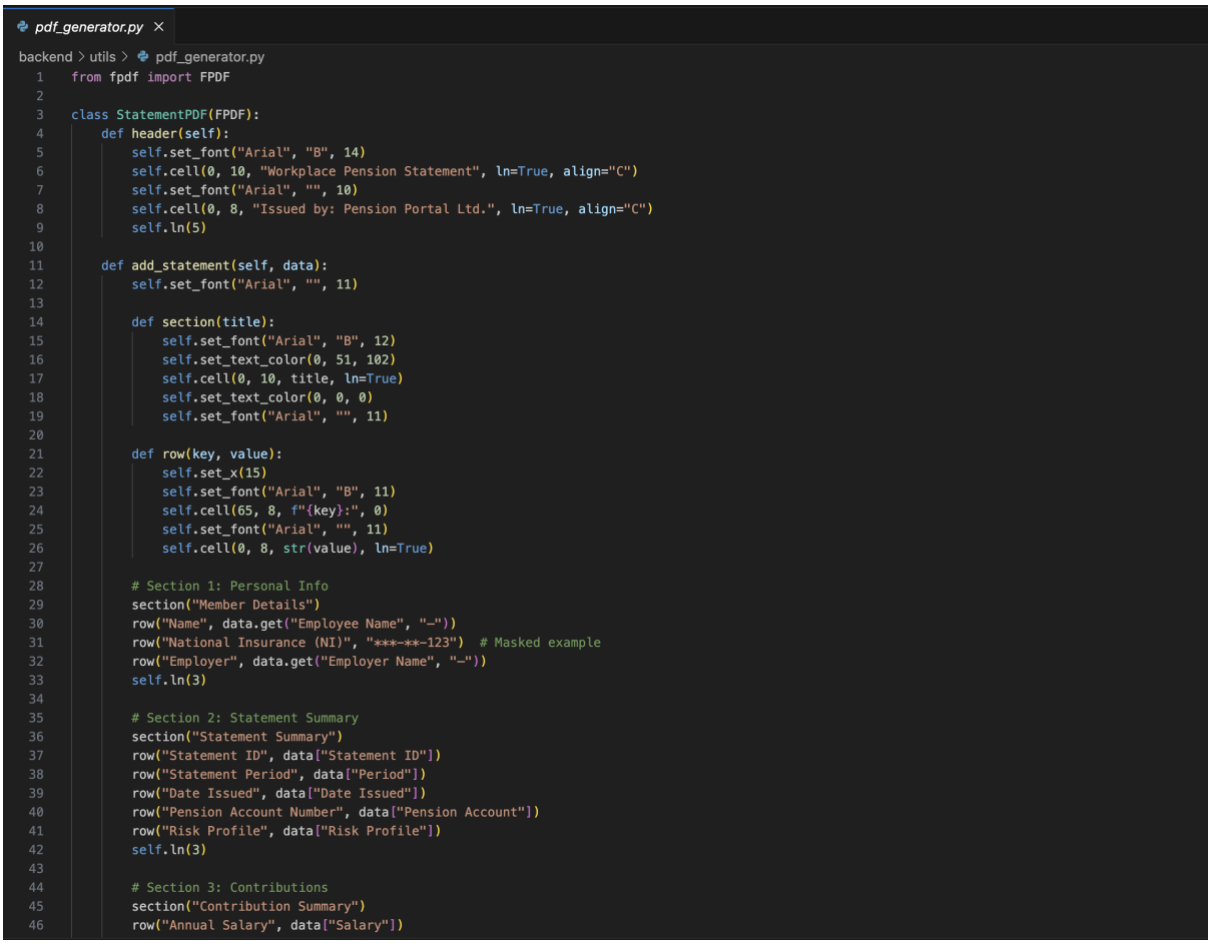
3. Secure PDF Generation of Pension Statements

File: pdf_generator.py

Purpose: Creates downloadable pension statements using user and fund data.

Key Features:

- Personalized PDF content per user
- Supports backend route for on-demand downloads



```python
from fpdf import FPDF

class StatementPDF(FPDF):
    def header(self):
        self.set_font("Arial", "B", 14)
        self.cell(0, 10, "Workplace Pension Statement", ln=True, align="C")
        self.set_font("Arial", "", 10)
        self.cell(0, 8, "Issued by: Pension Portal Ltd.", ln=True, align="C")
        self.ln(5)

    def add_statement(self, data):
        self.set_font("Arial", "", 11)

        def section(title):
            self.set_font("Arial", "B", 12)
            self.set_text_color(0, 51, 102)
            self.cell(0, 10, title, ln=True)
            self.set_text_color(0, 0, 0)
            self.set_font("Arial", "", 11)

        def row(key, value):
            self.set_x(15)
            self.set_font("Arial", "B", 11)
            self.cell(65, 8, f"{key}:", 0)
            self.set_font("Arial", "", 11)
            self.cell(0, 8, str(value), ln=True)

        # Section 1: Personal Info
        section("Member Details")
        row("Name", data.get("Employee Name", "—"))
        row("National Insurance (NI)", "***-**-123")  # Masked example
        row("Employer", data.get("Employer Name", "—"))
        self.ln(3)

        # Section 2: Statement Summary
        section("Statement Summary")
        row("Statement ID", data["Statement ID"])
        row("Statement Period", data["Period"])
        row("Date Issued", data["Date Issued"])
        row("Pension Account Number", data["Pension Account"])
        row("Risk Profile", data["Risk Profile"])
        self.ln(3)

        # Section 3: Contributions
        section("Contribution Summary")
        row("Annual Salary", data["Salary"])
```

Fig 8: Secure PDF Generation of Pension Statements

4. Role-Based Route Access Control
File: auth_middleware.py
Purpose: Restricts route access based on user role (admin, employer, employee).

Key Features:

- Centralized role enforcement across endpoints
- Prevents unauthorized access using JWT identity
- Enhances maintainability and security

auth_middleware.py

backend > utils > auth_middleware.py

```python
def token_required(f):
    @wraps(f)
    def decorated(*args, **kwargs):
        token = None

        # Extract token from header
        if 'Authorization' in request.headers:
            try:
                token = request.headers['Authorization'].split(" ")[1]
            except IndexError:
                return jsonify({'message': 'Invalid token format'}), 401

        if not token:
            return jsonify({'message': 'Token is missing!'}), 401

        try:
            data = jwt.decode(token, Config.JWT_SECRET_KEY, algorithms=["HS256"])

            current_user = None
            if data.get("role") == "admin":
                current_user = Admin.query.get(data.get("admin_id"))
            elif data.get("role") == "employer":
                current_user = Employer.query.get(data.get("employer_id"))
            elif data.get("role") == "user":
                current_user = User.query.get(data.get("user_id"))

            if not current_user:
                return jsonify({'message': 'User not found'}), 401

        except jwt.ExpiredSignatureError:
            return jsonify({'message': 'Token has expired!'}), 401
        except jwt.InvalidTokenError:
            return jsonify({'message': 'Invalid token!'}), 401
        except Exception as e:
            return jsonify({'message': str(e)}), 500

        return f(current_user, *args, **kwargs)

```

Fig 9: Role-Based Route Access Control

auth_middleware.py

backend > utils > auth_middleware.py

```python

def admin_required(f):
    @wraps(f)
    def decorated(*args, **kwargs):
        token = None

        if 'Authorization' in request.headers:
            token = request.headers['Authorization'].split(" ")[1]

        if not token:
            return jsonify({'error': 'Token is missing'}), 401

        try:
            data = jwt.decode(token, Config.JWT_SECRET_KEY, algorithms=["HS256"])

            # Check for 'admin' role from token payload
            if data.get('role') != 'admin':
                return jsonify({'error': 'Admin access only'}), 403

        except jwt.ExpiredSignatureError:
            return jsonify({'error': 'Token has expired'}), 401
        except jwt.InvalidTokenError:
            return jsonify({'error': 'Invalid token'}), 401

        return f(*args, **kwargs)
    return decorated

def employer_required(f):
    @wraps(f)
    def decorated(*args, **kwargs):
        token = None
        if 'Authorization' in request.headers:
            token = request.headers['Authorization'].split(" ")[1]

        if not token:
            return jsonify({'error': 'Token is missing'}), 401

        try:
            data = jwt.decode(token, Config.JWT_SECRET_KEY, algorithms=["HS256"])
            if data.get('role') != 'employer':
                return jsonify({'error': 'Employer access only'}), 403
        except jwt.ExpiredSignatureError:
            return jsonify({'error': 'Token expired'}), 401
        except jwt.InvalidTokenError:
            return jsonify({'error': 'Invalid token'}), 401
```

Fig 10: Role-Based Route Access Control

5. Frontend Multilingual UI Integration
File: home.component.html
Purpose: Supports dynamic content translation using ngx-translate.

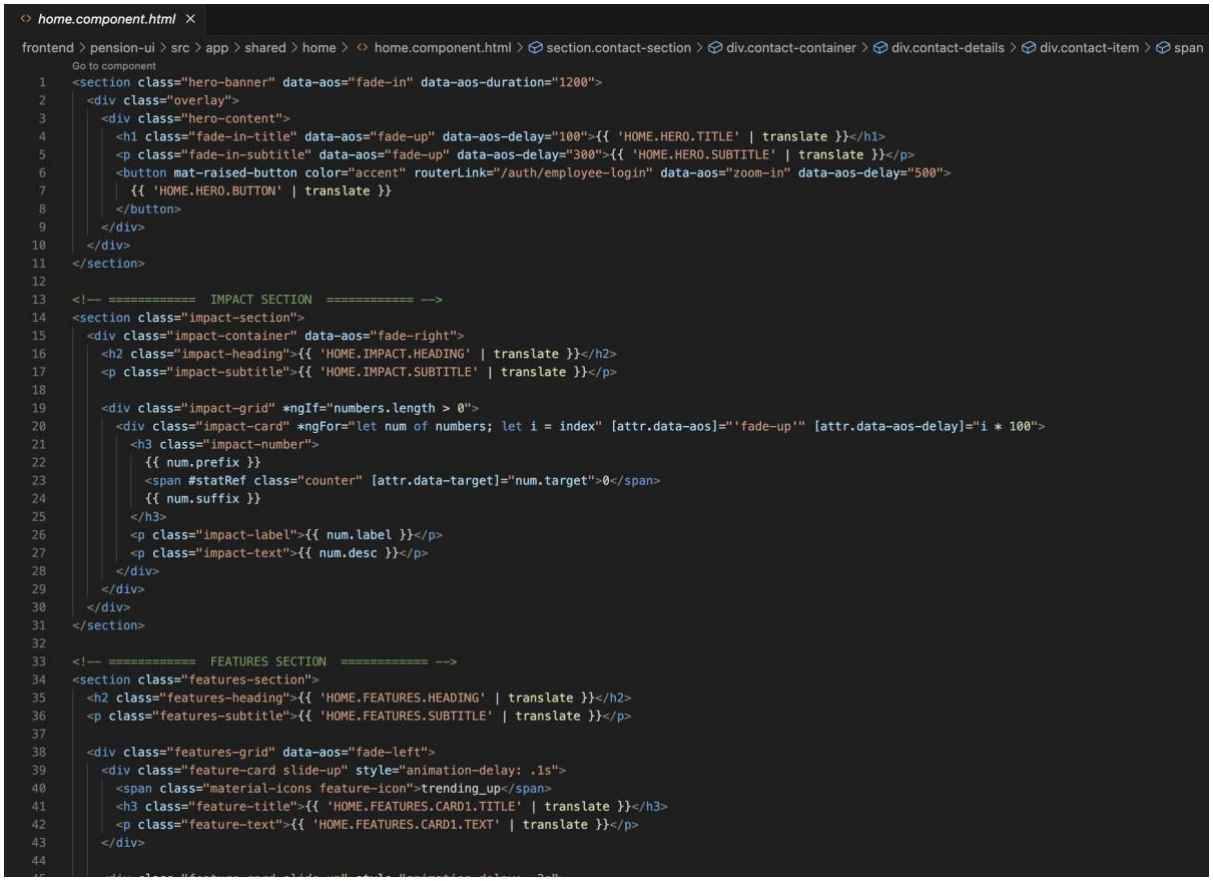


Fig 11: Frontend Multilingual UI Integration

Highlights:

- Uses en.json and hin.json for English and Hindi translations and similarly for Polish, French, German these other languages.
- Real-time language switching
- Fully integrated with Angular's i18n pipeline

6. Pension Analytics with Chart.js
File: analytics.component.ts
Purpose: Visualizes contribution trends using interactive line charts.

Highlights:

- Visual fund growth feedback for users
- Fully responsive across devices
- Enhances data transparency and engagement

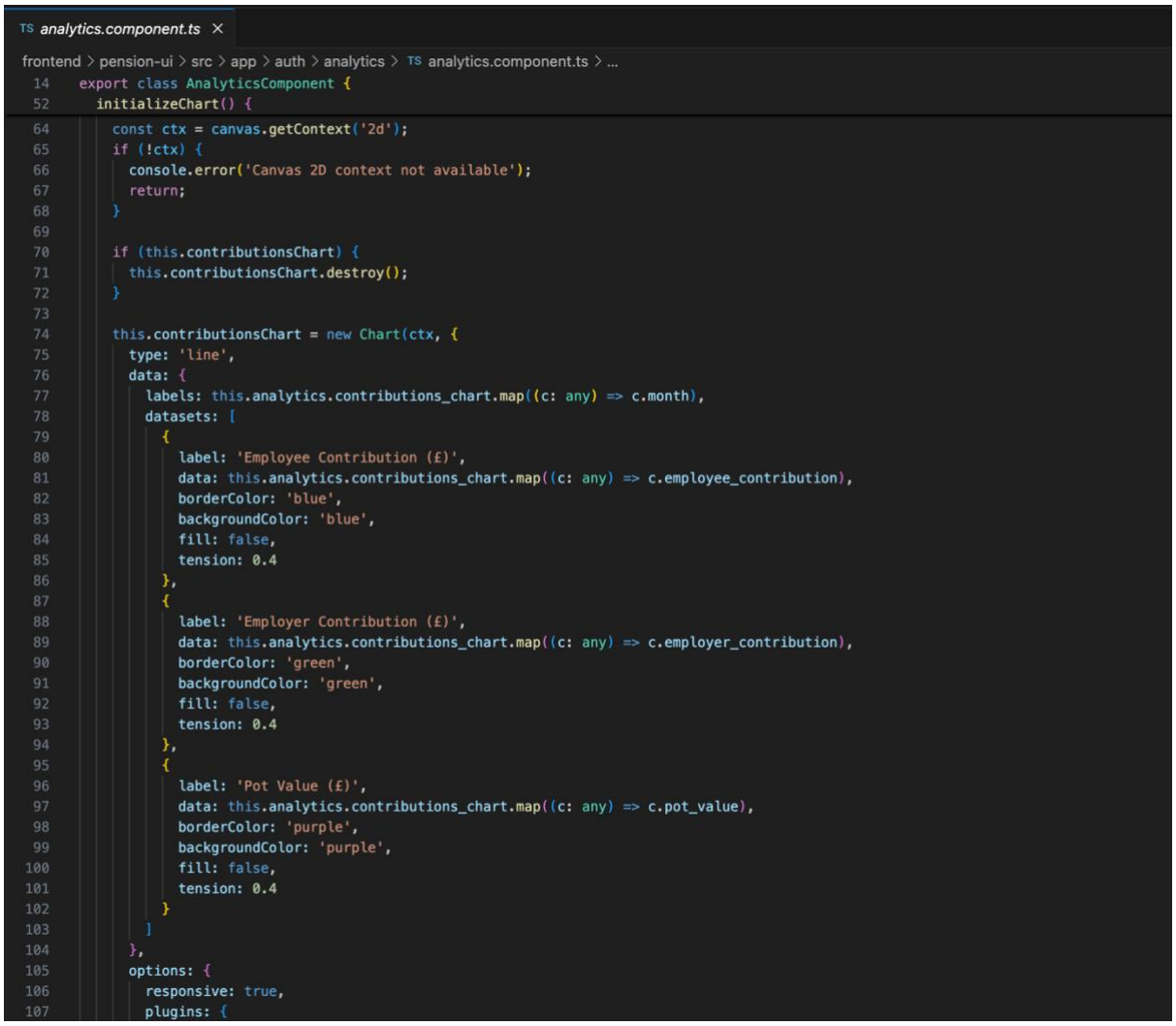


Fig 12: Pension Analytics with Chart.js

7. Voiceflow AI Chatbot Integration
File: voiceflow-loader.service.ts
Purpose: Provides conversational AI assistance for pension queries using Voiceflow.

Highlights:

- Embeds Voiceflow chatbot on all pages
- Handles user queries about pension statements, login issues, and UK pension policies
- Context-aware responses via integration with backend user session



Fig 13: Voiceflow AI Chatbot Integration

6. Testing and Evaluation

This section presents the testing strategy, tools, and outcomes used to validate the functional correctness, security, usability, and internationalization support of the pension management platform. Both automated API-level testing (via Postman) and manual UI testing were employed to ensure system robustness across different modules and user roles.

6.1 Postman API Testing

Postman was used to test all backend RESTful endpoints, covering authentication, role-based access, CRUD operations, and business logic. The testing involved creating and validating JSON Web Tokens (JWT), and verifying data integrity across multiple database entities.

- **Login with Correct Credentials & Valid MFA Token**

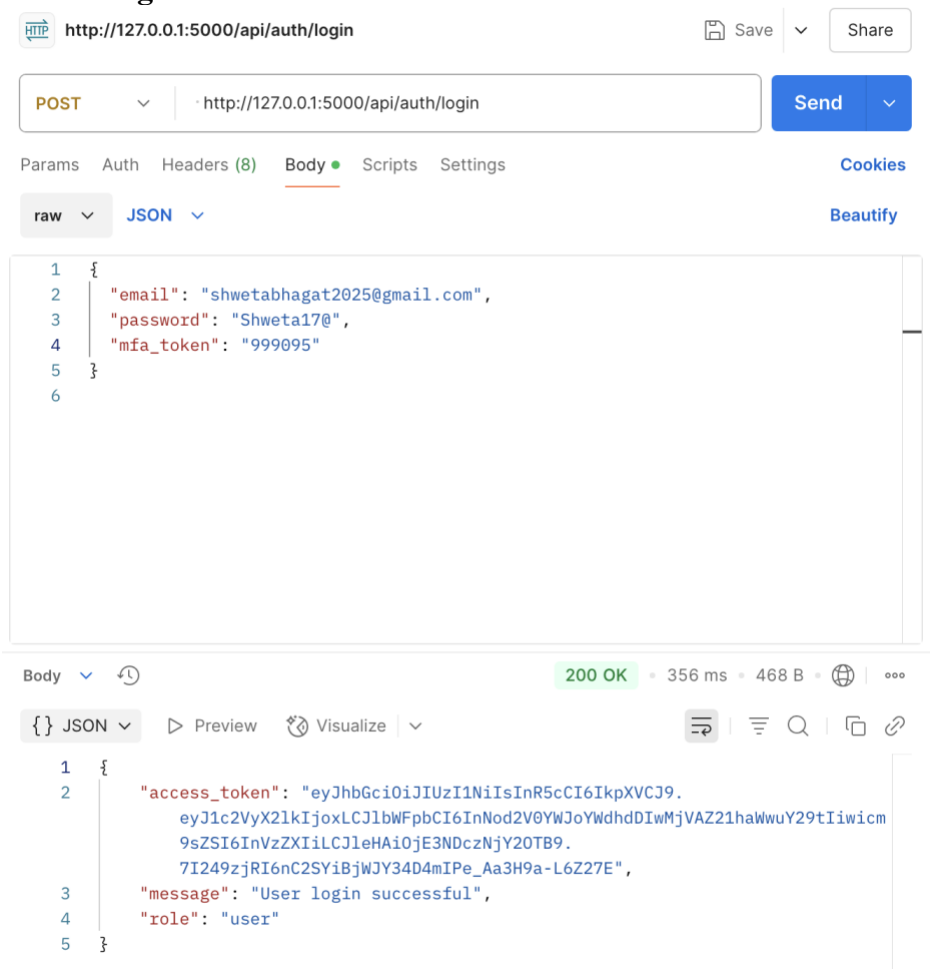


Fig 14: Login with Correct Credentials & Valid MFA Token

API successfully returns access token and confirms login.

- **Login with Invalid MFA Token**

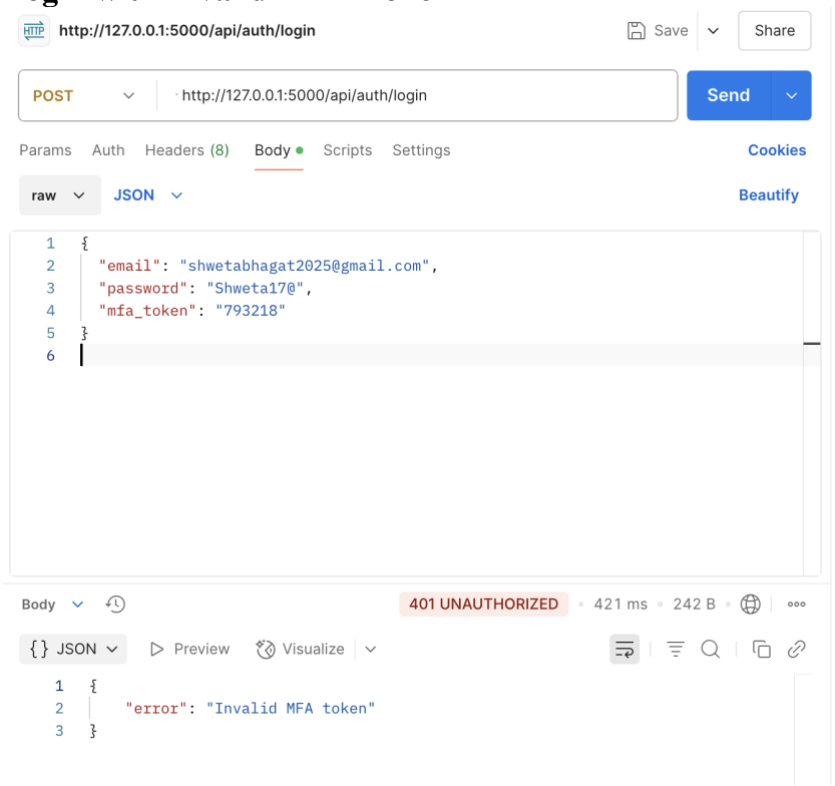


Fig 15: Login with Invalid MFA Token

Returns 401 Unauthorized with error: "Invalid MFA token".

- **Login with Incorrect Password**

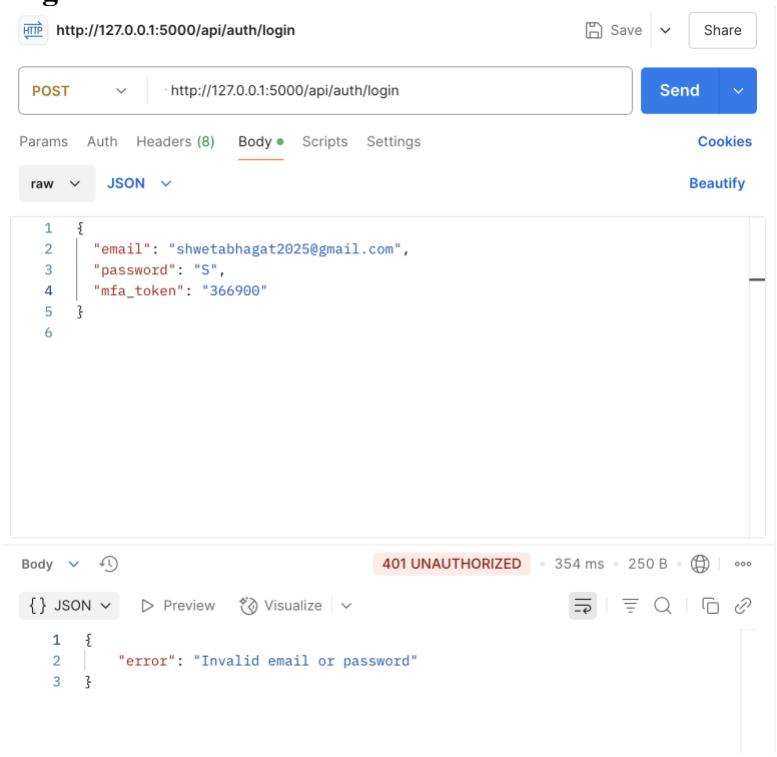


Fig 16: Login with Incorrect Password

Returns 401 Unauthorized with error: "Invalid email or password".

- **Access Beneficiary Endpoint with Invalid Token**

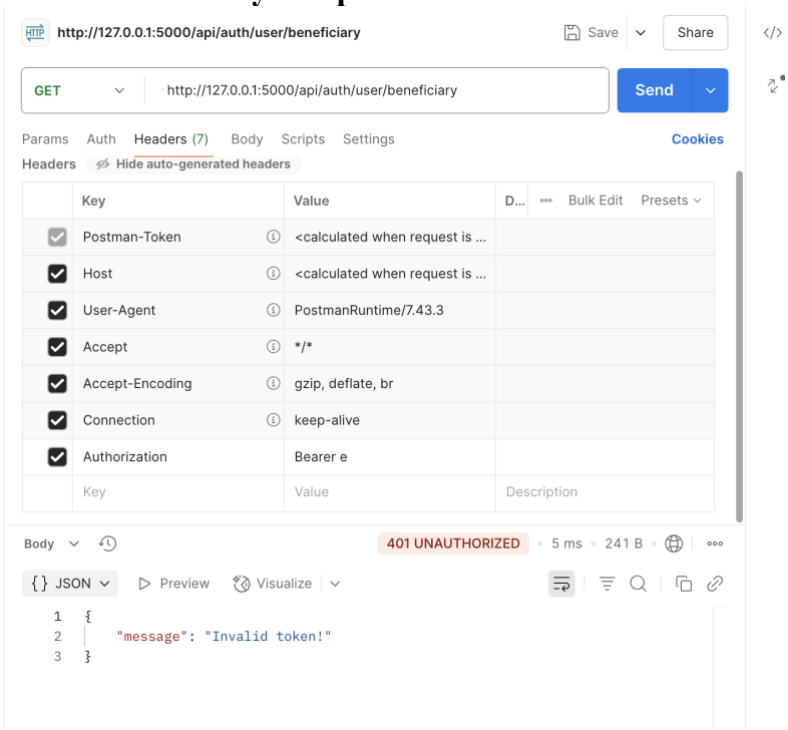


Fig 17: Access Beneficiary Endpoint with Invalid Token

Unauthorized access, token rejected.

- **Access Beneficiary Endpoint with Valid Token**

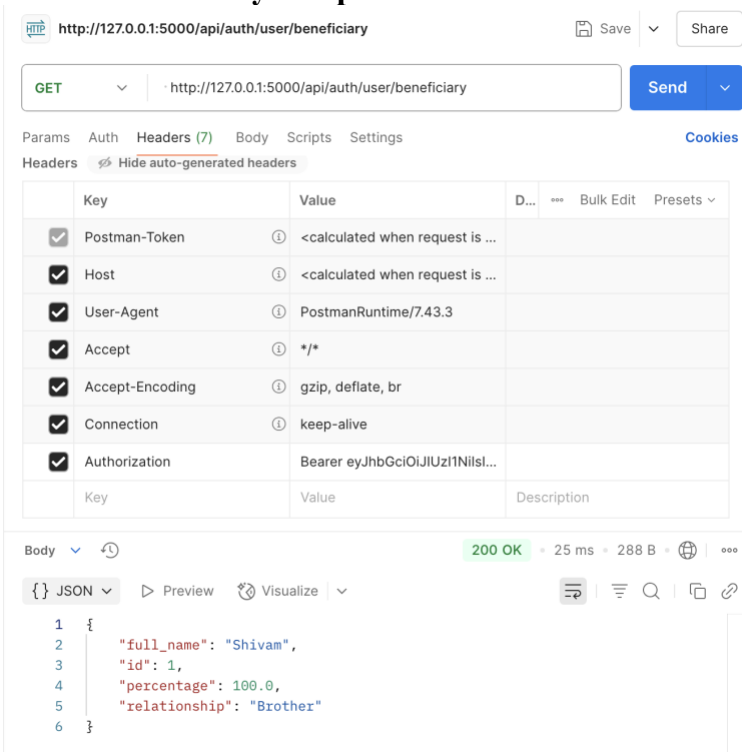


Fig 18: Access Beneficiary Endpoint with valid Token

Successfully retrieves beneficiary data

- **Logout with Invalid Token**

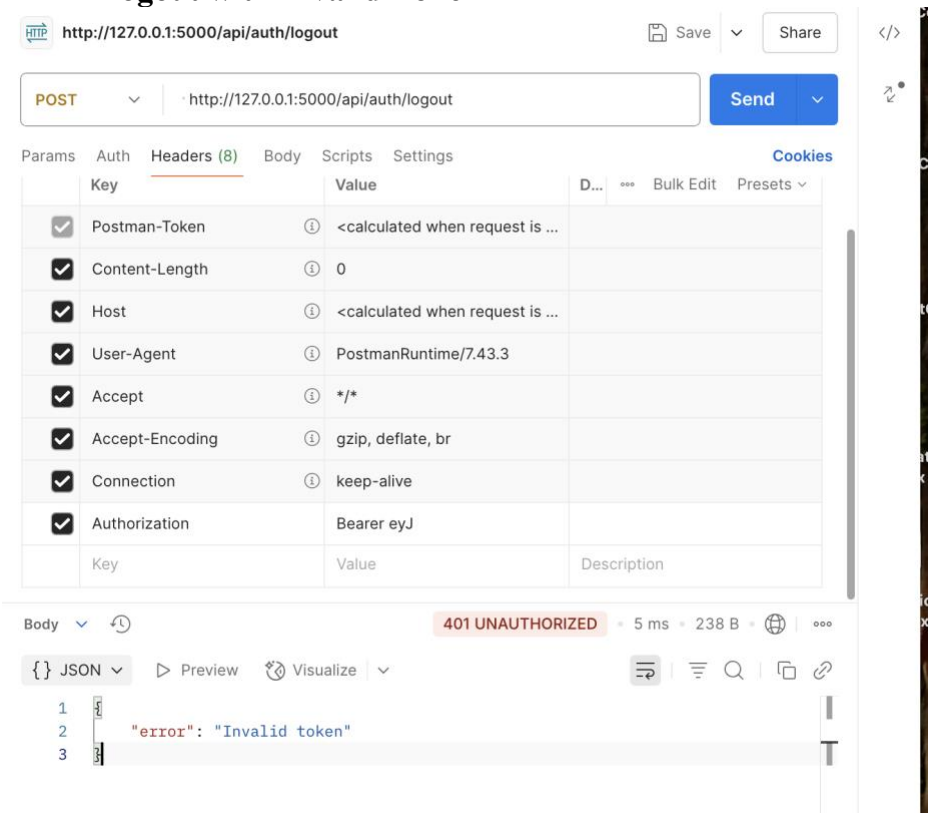


Fig 19: Logout with Invalid Token

Logout attempt fails with error: "Invalid token".

- **Logout with Valid Token**

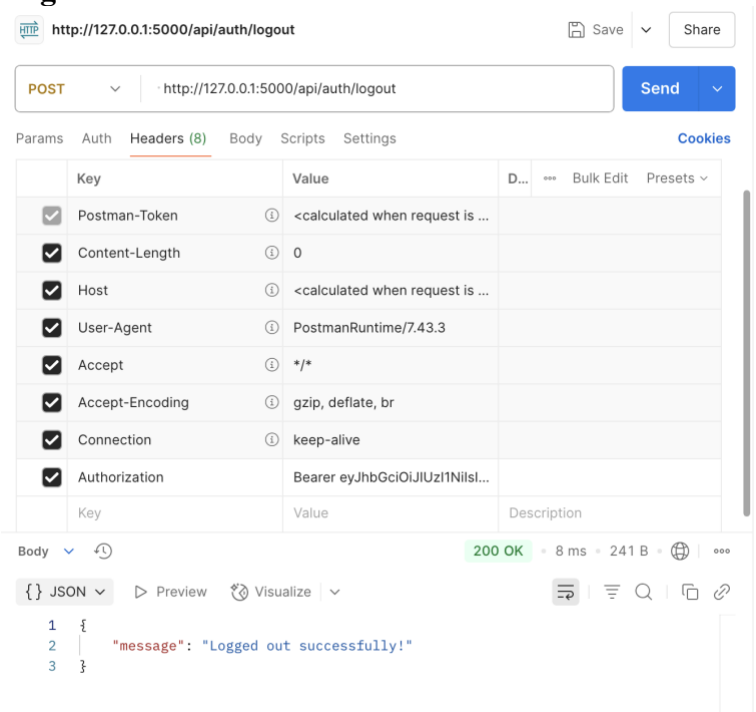


Fig 20: Logout with valid Token

Returns message: "Logged out successfully!

6.2 Integration Testing – Postman

Integration testing ensured that individual components work together as expected. This involved testing multi-step workflows using chained Postman requests and validating data flow between the frontend, backend, and database.

End-to-End Scenarios:

1. Admin adds employer → Employer adds employees → Statements generated → Employees view/download statements.
2. Password reset workflow via email token → Secure new password creation → Successful login with updated credentials.
3. Upload document → Store metadata → View/download file using user ID.

Results: All integrated workflows passed expected validations and state transitions.

6.3 Functional Testing – Postman

Functional testing focused on the correctness of core features and how the system responded to valid and invalid inputs. Each module was tested against its intended use cases.

Modules Tested:

- Login & logout sessions
- User registration & profile updates
- Document upload & retrieval
- Beneficiary management
- Statement generation and download
- Employer contribution upload

Validation Criteria:

- Correct HTTP status codes (200, 201, 400, 403, 404)
- JSON structure consistency
- Data persisted in MySQL
- Error handling for malformed input

Result: All modules functioned as expected, with error responses handled gracefully.

6.4 Security Testing – Postman

Security testing simulated real-world threats and verified the system's resilience against unauthorized access, token manipulation, and endpoint misuse.

Security Measures Tested:

Test Type	Description	Outcome
JWT Token Tampering	Modified payload and re-signed token	Access Denied (401)
Expired JWT Token	Access with expired session token	Session Rejected
Unauthorized Role Access	Employee accessing admin-only routes	Blocked (403)
IDOR Prevention	Employer accessing another employer's user data	Blocked
File Type Restriction	Uploading .exe disguised as .pdf	Rejected (400)
CSRF Simulation	Direct POST from another origin	Rejected (CORS)

Conclusion: Role-based restrictions, JWT handling, and file validation mechanisms were secure and properly enforced.

6.5 Internationalization (i18n) Testing

The platform supports English, Polish, French, German and Hindi using the ngx-translate library. Internationalization testing focused on the correct rendering of dynamic translations and fallback behaviour.

Tested Items:

- Static page content (Home, About, FAQ)
- Language switcher functionality
- Fallbacks for missing translation keys

Result:

- All translated content rendered correctly.
- Language toggling was real-time and state-preserving.
- No broken keys or untranslated strings in the UI.

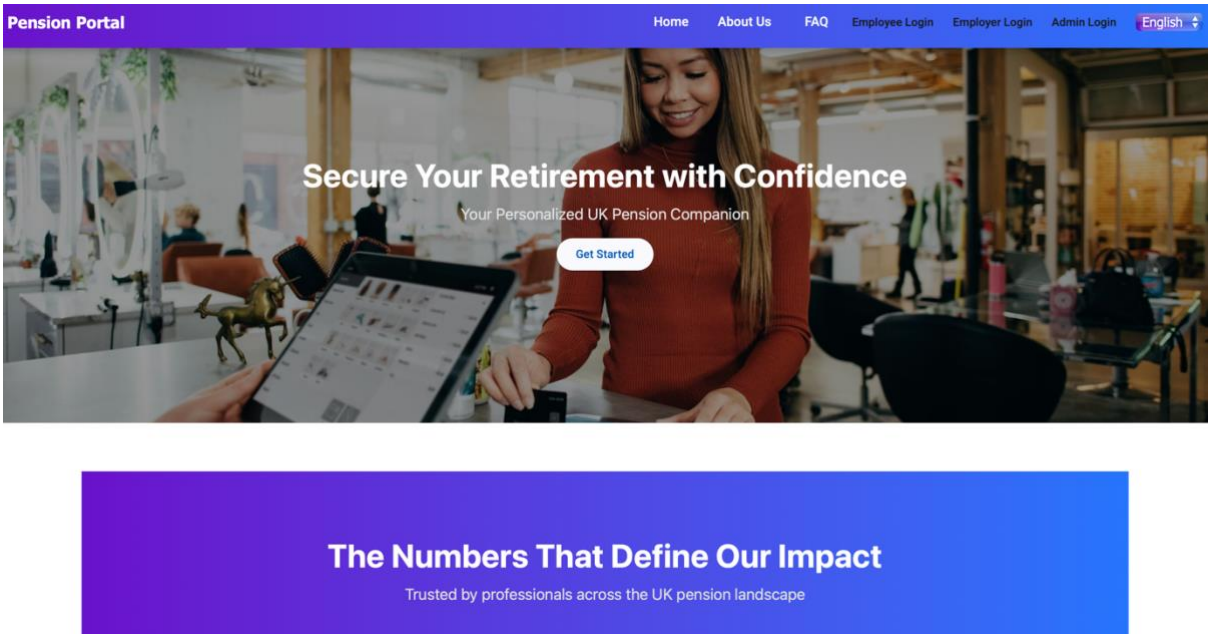


Fig 21: Home Page Multilanguage -English

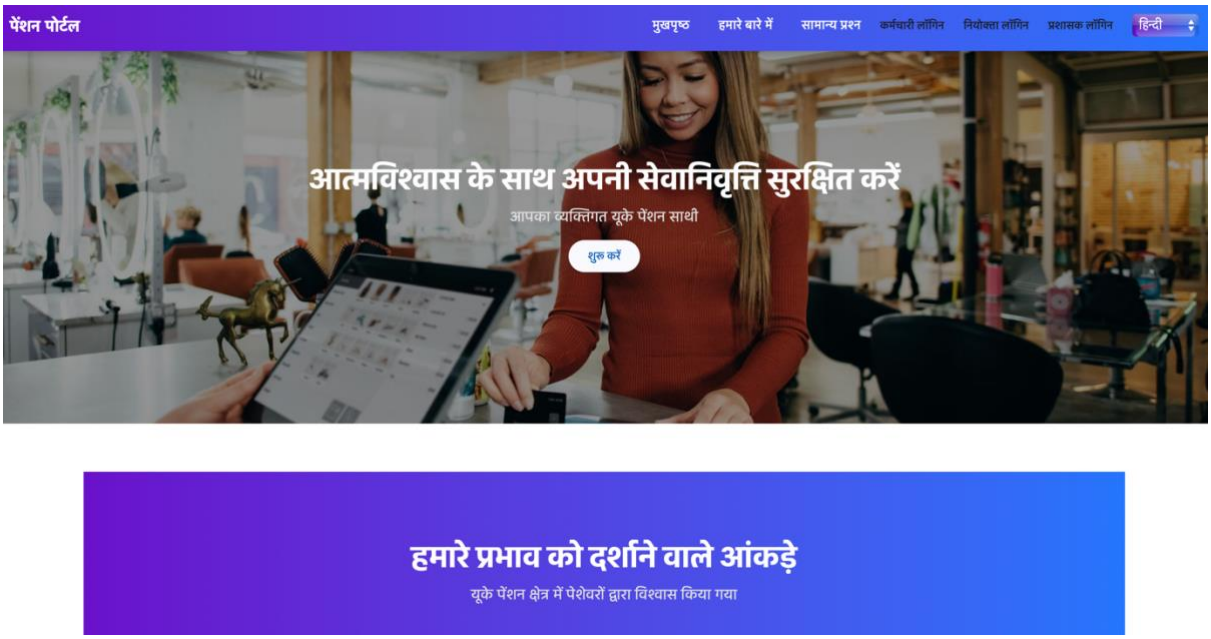


Fig 22: Home Page Multilanguage -Hindi

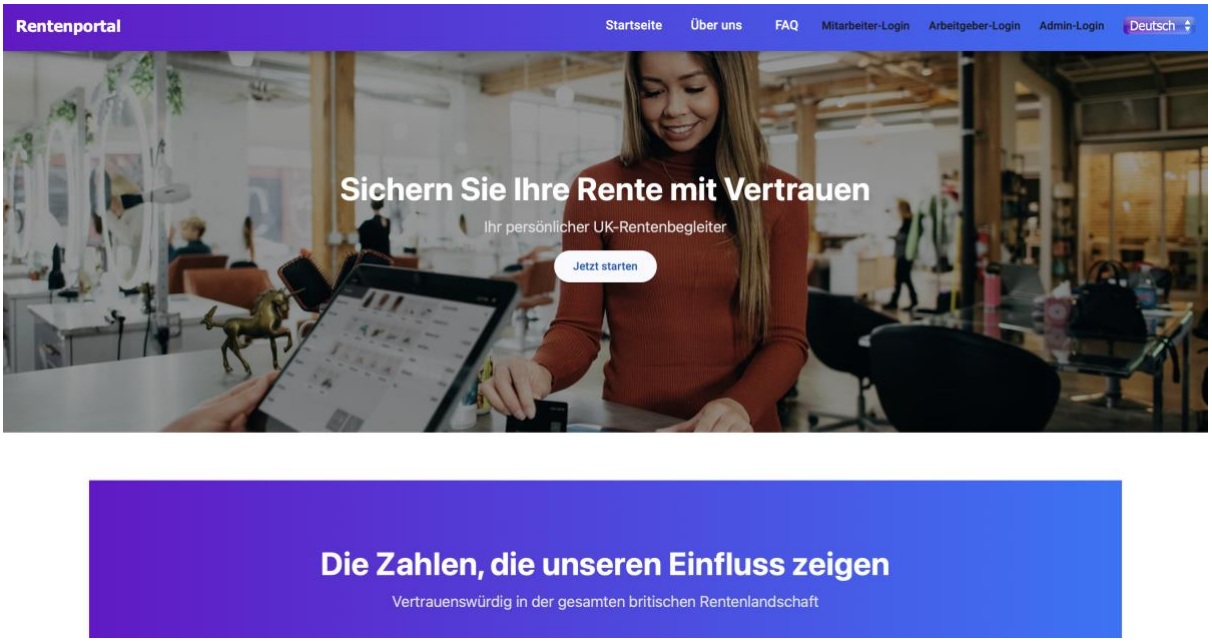


Fig 23: Home Page Multilanguage - German

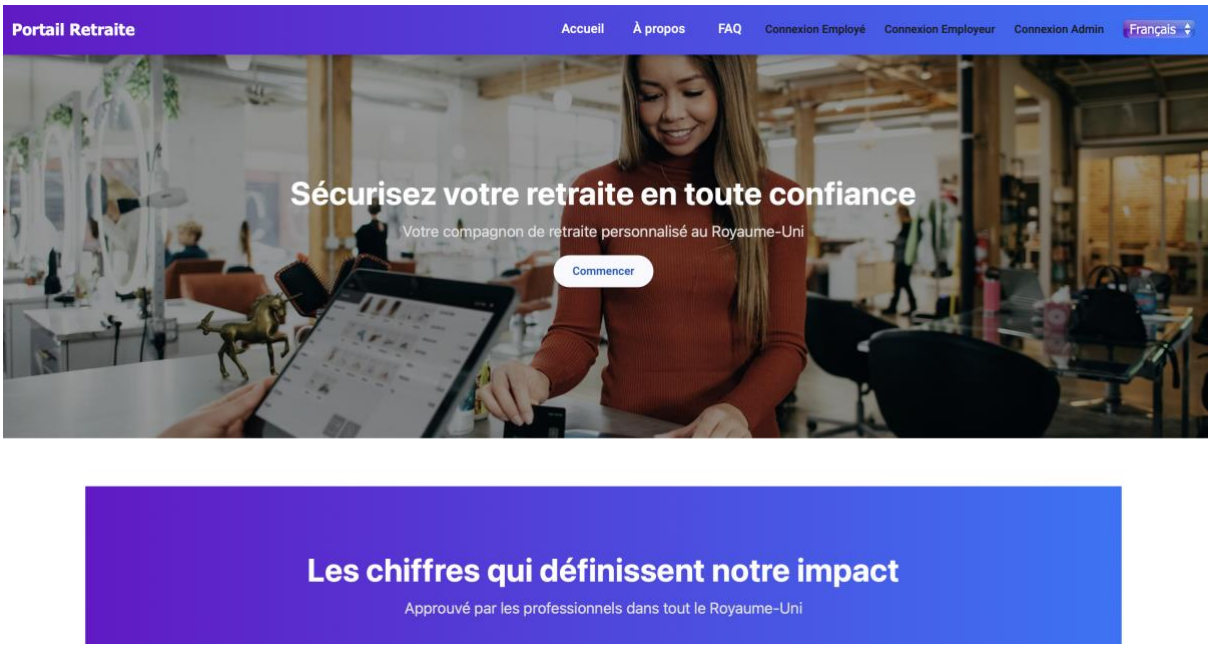


Fig 24: Home Page Multilanguage -French

6.6 Manual UI Testing

Manual UI testing was conducted across modern web browsers and device simulators to validate the interface's usability and consistency.

Browsers Tested:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari (macOS & iOS)

Tested UI Features:

Feature	Test Result
Form validations (empty/invalid fields)	Passed
Role-specific UI logic	Passed
Document viewer & download flow	Passed
Statement PDF links	Passed
Button/link accessibility	Passed
Navbar & footer behaviour	Passed
Font Awesome icons	Passed

Accessibility: Basic accessibility guidelines were followed, including appropriate colour contrasts, readable fonts, and button alt-labels.

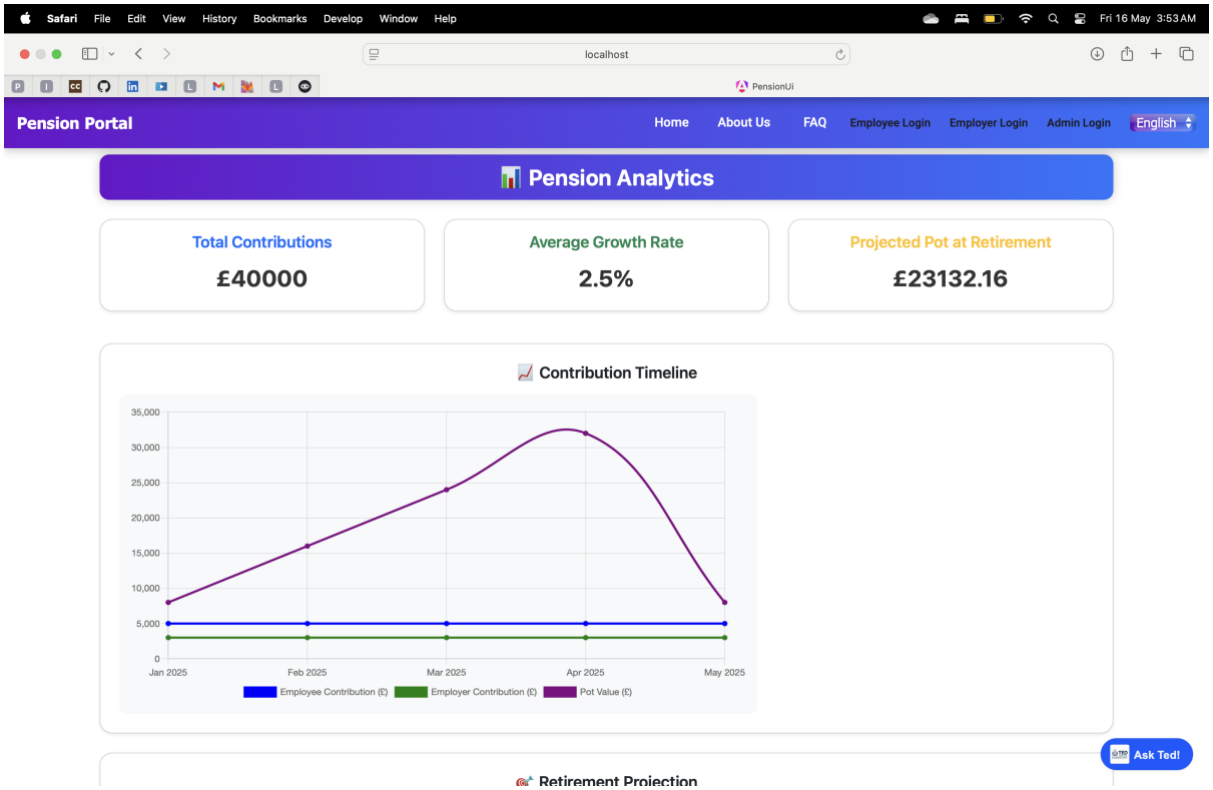


Fig 25: Safari Browser

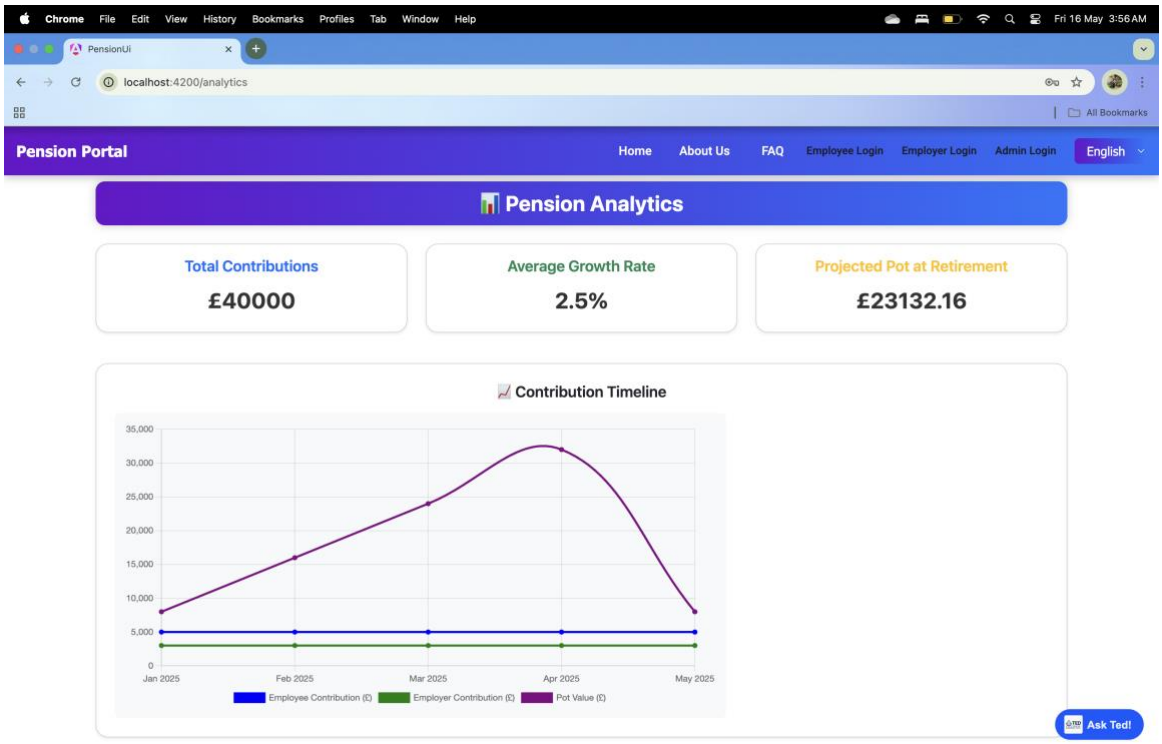


Fig 26: Chrome Browser

7. Critical Evaluation

7.1 Achievements

The pension management platform successfully fulfills its primary objectives of creating a secure, scalable, and user-friendly system for managing workplace pension schemes. Over the course of development, a range of technical, functional, and regulatory goals were met.

The following key achievements highlight the core accomplishments of the project:

1. Role-Based Access System

- A robust authentication and authorization system was implemented using **JWT tokens** and **Flask-Bcrypt**.
- **Multi-Factor Authentication (MFA)** was enforced for higher-privilege users (Admins and Employers), adding a vital security layer.
- Role-Based Access Control (RBAC) restricted features and views based on user roles (Admin, Employer, Employee).

2. Monthly Pension Statement Automation

- Backend logic automatically calculates monthly contributions (3% employer + 5% employee) and simulates fund growth.
- Statements are generated via scheduled tasks using **Flask-APScheduler**.
- Statements are viewable on the user dashboard and downloadable in **PDF format**, ensuring transparency.

3. Modern Frontend (Angular)

- The Angular-based UI is **modular, clean**, designed with Bootstrap for multi-device support.
- Navigation and component logic adapt based on user roles, with intuitive dashboards for each.
- Integrated **Chart.js** visualizations allow employees to view pension growth trends graphically.

4. Multilingual Interface

- The application supports **multi-language switching (English, Polish, French, German and Hindi)** via ngx-translate.
- Language support has been applied across static content including homepage, about page, FAQs, and login pages.
- Enables better accessibility for users from diverse backgrounds.

5. Voiceflow AI Chatbot Integration

- A **Voiceflow-powered chatbot** is embedded to help users with common pension-related queries.
- The chatbot improves user support without relying on manual intervention.
- Future extensibility allows linking it to backend data (e.g., fetching pension dates, FAQs, etc.).

6. Secure Document Handling

- Employees can **upload, view, and download** pension-related documents.
- All documents are associated with user IDs and stored securely on the backend, simulating real-world compliance processes.

7. Employer Portal Features

- Employers can **add/view employees**, manage contribution uploads, and access statements.
- Employer profiles support secure password resets and MFA, with limited views restricted to their associated users.

8. Admin Dashboard & System Control

- Admins can **add/delete employers**, view all users in the system, and manage global system configurations.
- Admins can also trigger statement generation and manage data integrity at a system-wide level.

9. GDPR-Inspired Compliance Measures

- Data access is protected using secure tokens, hashed credentials, and encrypted API routes (HTTPS).
- Users can **update or delete personal information**, simulating GDPR compliance workflows.
- Sessions are token-based with blacklist support for secure logout and revocation.

10. Scalability & Maintainability

- Modular folder structure and clean separation of concerns between frontend, backend, and database logic.
- Uses version control (GitLab) and Flask-Migrate for future-proof schema evolution.

This set of achievements reflects a well-rounded, full-stack software development effort that meets the functional and security demands of a workplace pension system, while also integrating cutting-edge usability features like multilingual support and AI assistance.

7.2 Challenges and Solutions

Throughout the development of the pension management platform, several technical, architectural, and design challenges were encountered. These obstacles required thoughtful problem-solving and iterative improvements to ensure the system remained robust, user-friendly, and aligned with project objectives. The following outlines key challenges and the solutions adopted:

1. Challenge: Implementing Role-Based Access Across Multiple User Types
Description: Managing authentication and authorization for multiple user roles (Admin, Employer, Employee) without creating code duplication or access vulnerabilities was complex.
Solution: A centralized Role-Based Access Control (RBAC) middleware was developed using Flask decorators. This middleware validated JWTs and verified the user's role before allowing access to protected routes. Angular routing also mirrored these restrictions to ensure frontend-level access control.

2. Challenge: Ensuring Multi-Factor Authentication (MFA)
Description: Integrating MFA for Admins and Employers required secure handling of time-sensitive tokens while maintaining a smooth login experience.
Solution: A TOTP-based MFA system was implemented using time-based one-time password logic, synced with the user's secret key stored in the database. MFA token validation was added to the login flow for higher-privilege roles without affecting standard employee login.

3. Challenge: Monthly Statement Automation
Description: Automatically generating personalized pension statements each month for every employee required coordination between scheduling, financial calculations, and data persistence.
Solution: A Flask-APScheduler job was configured to run monthly, dynamically calculating contributions and fund growth. Statements were created using ORM models and rendered into downloadable PDFs using ReportLab.

4. Challenge: Multilingual UI Integration
Description: Angular's internationalization required all static content to be restructured and translated, posing challenges for consistency and maintainability.
Solution: The ngx-translate library was used to manage translations. JSON files (en.json, hin.json, pl.json, etc) were created to store key-value translations, and all static content was updated with the translate pipe in Angular templates.

5. Challenge: Designing for Diverse User Experiences
Description: Catering to both high-privilege users (Admins, Employers) and regular employees with different workflows increased UI complexity.
Solution: Dedicated dashboard components were developed for each role, ensuring UI clarity and logical separation. Angular route guards and services helped manage session flow and user redirection based on roles.

6. Challenge: Embedding and Securing Chatbot Integration
Description: The integration of the Voiceflow chatbot posed security and performance concerns related to external script loading and session context handling.
Solution: The chatbot was embedded securely in the index.html with scoped loading, and future enhancements were planned to pass user metadata securely via backend proxy if deeper personalization is required.

7. Challenge: Secure File Upload and Download

Description: Allowing users to upload pension documents while preventing unauthorized access or malicious uploads required strict validation and access control.
Solution: Upload routes validate file types and sanitize filenames. Each document is stored with user ID mapping, and download endpoints are protected by JWT tokens and RBAC checks.

8. Challenge: Managing Data Integrity Across Relationships
Description: Maintaining referential integrity between users, employers, beneficiaries, and pension statements required careful database design and migration planning.
Solution: The schema was normalized, with foreign key constraints enforced using Flask-SQLAlchemy relationships. All deletions and updates were designed with cascading behaviour or restricted actions to preserve integrity.

9. Challenge: Hosting & Performance
Description: Simulating production performance locally while ensuring API response times remained below 2 seconds.
Solution: API endpoints were optimized using indexed queries and pagination (for future scaling). Local testing included concurrent session simulations and PDF generation load testing.

These challenges reflect the complexity of building a real-world, secure financial platform. By applying modular design principles, adopting robust libraries, and following best practices in authentication and architecture, the project was able to overcome these difficulties and meet its goals effectively

7.1 Limitations and Final Evaluation

Despite the successful implementation of a comprehensive pension management system, a few limitations were observed that present opportunities for future improvement. These limitations primarily stem from resource constraints, time boundaries, and the simulated nature of the project.

1. Limited Financial Complexity
Limitation: The system currently applies a fixed contribution percentage (3% employer, 5% employee) and a static fund performance rate for pension growth simulation. This does not reflect the complex variability found in real-world pension funds, which depend on market conditions, fund selection, and economic indicators.
Evaluation: While sufficient for simulation, future iterations should integrate real-time financial APIs or allow users to select from dynamic investment portfolios with historical growth trends.

2. No Real Banking or Payroll Integration
Limitation: The platform does not integrate with real payroll or banking systems to reflect actual salary deductions or fund transfers.

Evaluation: This decision was made to maintain the academic scope and security of the project. However, integrating with banking APIs or sandbox environments could elevate the realism and applicability of the system in professional deployments.

3.Basic AI Chatbot Interaction

Limitation: Although a Voiceflow chatbot is embedded and functional, it currently relies on pre-trained responses and lacks personalized financial advice or data-driven context.
Evaluation: Future improvements could include backend-connected AI that can fetch real-time user data (e.g., pension pot value or contribution dates) to respond more intelligently to queries.

4. Limited Language Support

Limitation: Only five languages English, Polish, French, German and Hindi are supported through the multi-language interface.
Evaluation: Adding more languages and storing user language preferences could significantly enhance the accessibility of the platform for the UK's diverse workforce.

5. Static PDF Layouts

Limitation: PDF statements are functional but visually basic, lacking branding, graphical fund illustrations, or financial disclaimers often seen in real pension reports.
Evaluation: Integrating a template-based PDF generator or adopting styled PDF libraries could improve presentation quality for professional use cases.

6. Manual MFA Setup

Limitation: The current MFA setup does not include a QR code onboarding or time-limited backup codes, which are common in production-grade systems.
Evaluation: Introducing QR-based MFA setup (e.g., via Google Authenticator integration) and fallback options would enhance usability without compromising security.

8. Future Work

While the current version of the pension management platform successfully meets the core objectives of security, usability, and regulatory alignment, several opportunities exist to enhance its functionality, realism, and impact in real-world scenarios. Future work can focus on expanding system capabilities, improving user experience, and deepening financial intelligence through smart automation and personalization.

1. AI-Powered Financial Advisor Integration
To make the platform more intelligent and user-centric, an AI-based financial advisor could be integrated. This system could analyse user contribution history, salary patterns, and fund performance to provide:

- Personalized retirement savings forecasts.
- Recommendations on increasing contributions.
- Fund allocation suggestions based on investment risk appetite (low/medium/high). This could be implemented using machine learning libraries such as TensorFlow or scikit-learn, integrated with user data through secure APIs.

2. Multi-Language Expansion
Currently, the platform supports English, Polish, French, German and Hindi. As pension tools cater to a diverse population in the UK, future work could expand support for additional languages. This would make the platform more inclusive and accessible to non-English speaking users across workplaces.

3. Real-Time Investment Data Integration
To simulate real fund performance more accurately, integration with live financial APIs (e.g., Yahoo Finance, Alpha Vantage) can be considered. This would allow:

- Realistic simulations of pension fund growth based on market trends.
- Dynamic updates to users' pension pot value and projections.

4. Payroll and Banking Integration
Integrating the system with payroll software or banking APIs (such as Plaid or Open Banking APIs) could allow:

- Automatic syncing of real salary contributions.
- Direct fund transfers and deductions.
- Real-time verification of contribution receipts.

5. Mobile App Version
While the current platform is mobile-responsive, a native mobile app could further improve accessibility. Future development could include:

- Flutter or React Native-based mobile applications.
- Push notifications for statement availability, contribution changes, or security alerts.

6. Automated Statement Email Delivery

Currently, pension statements are only available via dashboard download. An optional feature to **automatically email monthly/annual statements** securely to users would improve convenience and simulate real-world pension delivery services.

7. Enhanced Reporting and Fund Switch Feature
Admins or employers could be provided with advanced analytics features, such as:

- Cross-employee contribution summaries.
- Fund switch or reallocation functionality for employees based on new fund options or risk categories.

8. Pension Education and Gamification
To improve financial literacy and user engagement, gamification features can be added:

- Learning modules about pension types and retirement options.
- Progress badges and streaks for active contribution or profile completion.
- Simulated retirement scenarios to help users plan better.

9. Advanced Security Measures
While MFA and JWT are in place, future iterations could include:

- **Biometric login support** for mobile apps.
- **Session expiration alerts**.
- **Audit logging dashboard** for Admins to monitor suspicious activities.

10. Regulatory Reporting and Compliance Tools
To align with evolving pension regulations, the platform could support:

- Auto-generation of compliance reports for regulators.
- Secure audit logs accessible by authorized pension officers.
- Automatic notification if contribution percentages fall below the UK statutory minimum.

These future work suggestions aim to extend the scope of the current platform from a simulation tool to a near-production-ready financial management solution. They also demonstrate the platform's flexibility and potential to evolve in alignment with digital transformation trends in the fintech and pension sectors. With strategic upgrades, the platform can be transformed into a comprehensive digital pension service catering to both enterprise and individual needs.

9. Conclusion

The successful development of this pension management platform marks a significant step toward modernizing and simplifying workplace pension processes. By simulating a UK-compliant pension scheme, the system provides a secure, user-friendly, and scalable solution for employees, employers, and administrators to manage retirement contributions, access pension statements, and engage with long-term financial planning.

The project achieved its core objectives, including the implementation of secure user authentication (with MFA), role-based access control (RBAC), real-time pension tracking, and dynamic fund visualization. Through the use of modern technologies such as Angular, Flask, MySQL, and Chart.js, combined with PDF generation and multilingual support, the platform offers a rich feature set that mirrors the essential functions of real-world pension systems like “NEST” and “The People’s Pension”.

Additionally, the integration of a Voiceflow-powered AI chatbot and support for multiple languages (English, Polish, French, German and Hindi) enhances the inclusivity and accessibility of the platform providing a practical demonstration of how fintech applications can be extended for wider adoption. Each module of the system from authentication to document handling was developed with a strong emphasis on data security, regulatory compliance (GDPR, TPR), and user experience.

Throughout the development process, challenges such as managing multiple user roles, automating monthly statement generation, and implementing secure PDF handling were addressed with practical and scalable solutions. While the platform does simulate a workplace pension system effectively, several limitations including simplified fund simulation, lack of real-time payroll integration, and basic chatbot logic present avenues for future enhancement.

In summary, the project not only delivers a functional and technically sound pension management tool but also serves as a prototype for real-world applications in the financial services sector. It demonstrates how emerging developers can apply full-stack development practices, secure coding principles, and user-centred design to address meaningful societal needs such as retirement planning and financial literacy.
This work lays a solid foundation for future innovation and serves as a valuable portfolio piece that reflects both technical capability and problem-solving skills in a critical domain of financial technology.

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